

Requirement & Design Specification

PhucNguyen Resort & Tour Hub

Version: 1.0

– Hanoi, June 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Overall Requirements

1.1 Context Diagram

PhucNguyen Resort & Tour Hub is a web-based integrated resort management system that combines hotel booking, restaurant services, tour booking, billing, and operational management into a single platform. The system enables guests to conveniently access resort services while helping staff and managers efficiently manage daily operations, payments, customer information, and business performance through centralized dashboards and reports.

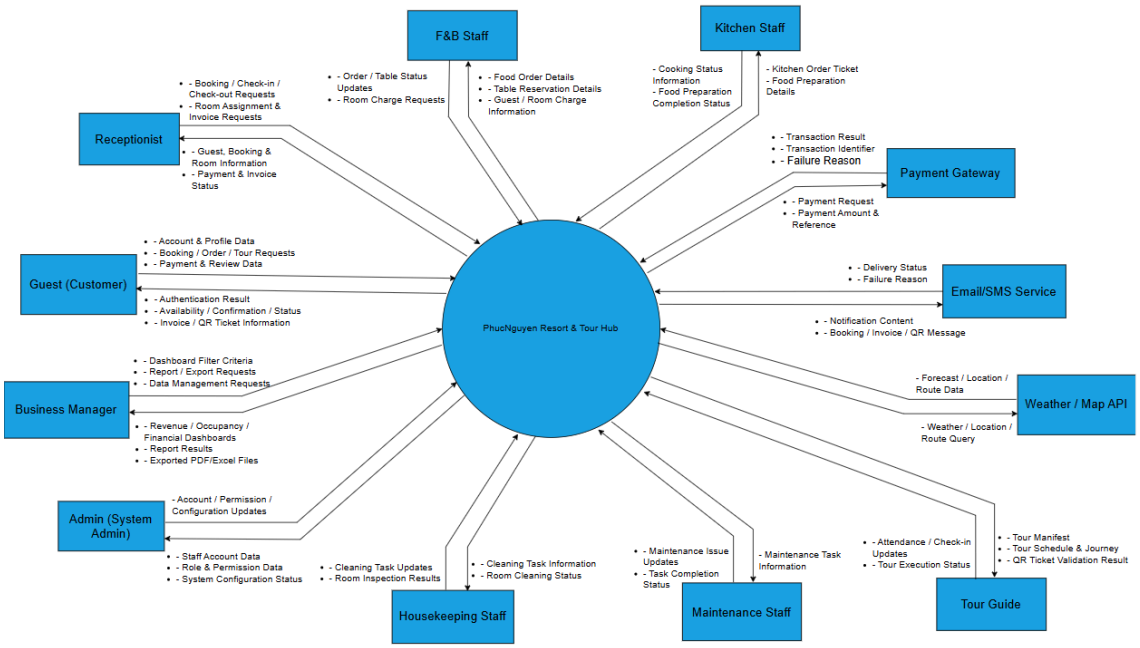


Figure I.1: Context Diagram

[Context Diagram]

[Draw.io]

The system is developed using the following technologies:

- Programming Language: C#
- Framework: ASP.NET Core MVC .net8
- Database:SQL Server

From context diagram to use case modeling

1.2 User Requirements

1.2.1 Actors

#	Actor	Description
1	Guest	An unauthenticated user who accesses the system without logging in. Guests can browse resort information, search available rooms, view restaurant menus, explore tour packages, read customer reviews, and register for an account.
2	User	The user represents all external entities, specifically human beings. Users can log in, manage their personal profile, update personal information, and access authenticated system features.
3	Customer	A registered and authenticated user who actively uses resort services. Customers can book rooms, make deposit payments, reserve restaurant tables, order food, book tours, view invoices, submit reviews, and request personal data deletion.
4	Admin	The system administrator with the highest level of authority. Responsible for managing the entire system, user accounts, role assignments, account

		suspension/reactivation, data backup and recovery, and monitoring system activities.
5	Business Manager	The resort operations manager responsible for managing business-related data such as room categories, restaurant menus, and tour packages. The manager can monitor revenue, occupancy rates, financial reports, customer reviews, and overall business performance.
6	Receptionist	Front desk staff responsible for managing reservations, verifying guest information, performing check-in and check-out procedures, assigning rooms, updating room statuses, processing final payments, and generating consolidated invoices.
7	Housekeeping Staff	Staff responsible for maintaining room cleanliness. They can view cleaning tasks, update room cleaning statuses, mark rooms as cleaned, and report maintenance issues when necessary.
8	Maintenance Staff	Technical staff responsible for inspecting, repairing, and maintaining resort facilities and equipment. They manage maintenance requests, perform repairs, and update maintenance task statuses.
9	F&B Staff	Restaurant and room service staff responsible for handling food orders, managing table reservations, updating order statuses, delivering food, and charging food expenses to guest room accounts.
10	Kitchen Staff	Kitchen personnel responsible for receiving Kitchen Order Tickets (KOTs), preparing food, updating food preparation statuses, and marking orders as ready for service.

1.2.2 Use Cases Description (UC)

ID	Use Case	Actor(s)	Use Case Description
UC01	Login	Guest	As a guest, I want to log into the system so that I can access features according to my role.
UC02	View Homepage	Guest	As a guest, I want to view the homepage so that I can access resort information and available services.
UC03	View Rooms	Guest	As a guest, I want to browse available rooms so that I can find suitable accommodation.
UC04	Search Room	Guest	As a guest, I want to search rooms based on my preferences so that I can find suitable rooms quickly.
UC05	Filter Room	Guest	As a guest, I want to filter rooms by category, price, and capacity so that I can narrow down my options.
UC06	View Room Details	Guest	As a guest, I want to view room details so that I can make an informed booking decision.
UC07	View Restaurant Table	Guest	As a guest, I want to view restaurant tables so that I can reserve a table if needed.
UC08	View Table Details	Guest	As a guest, I want to view table details so that I can select a suitable table.

UC09	View Restaurant Menu	Guest	As a guest, I want to browse the restaurant menu so that I can explore available dishes.
UC10	View Food Details	Guest	As a guest, I want to view food details so that I can know ingredients, prices, and descriptions.
UC11	View Tours	Guest	As a guest, I want to browse available tours so that I can plan activities during my stay.
UC12	View Tour Details	Guest	As a guest, I want to view detailed tour information so that I can decide whether to book it.
UC13	View Resort Information	Guest	As a guest, I want to view resort information so that I can learn about facilities and services.
UC14	View Reviews	Guest	As a guest, I want to view customer reviews so that I can evaluate service quality.
UC15	View Profile	Registered User	As a registered user, I want to view my profile information so that I can verify my account details.
UC16	Update Profile	Registered User	As a registered user, I want to update my profile information so that my account information remains accurate.
UC17	Update CCCD/Passport	Registered User	As a registered user, I want to upload or update my identification documents for verification purposes.
UC18	Change Password	Registered User	As a registered user, I want to change my password so that I can secure my account.
UC19	Forgot Password	Registered User	As a registered user, I want to reset my password when I forget it so that I can regain access to my account.
UC20	Logout	Registered User	As a registered user, I want to log out of the system so that my account remains secure.
UC21	Register Account	Customer	As a customer, I want to create an account so that I can use resort services.
UC22	Book Room	Customer	As a customer, I want to book a room so that I can stay at the resort.
UC23	Payment	Customer	As a customer, I want to pay deposits and service fees online so that my bookings can be confirmed.
UC24	Reserve Restaurant Table	Customer	As a customer, I want to reserve a restaurant table so that I can dine at my preferred time.
UC25	Order Food To Room	Customer	As a customer, I want to order food to my room so that I can enjoy room service.
UC26	Book Tour	Customer	As a customer, I want to book a tour so that I can participate in resort activities.
UC27	View Personal Invoice	Customer	As a customer, I want to view my invoices so that I can track my expenses.
UC28	View Booking History	Customer	As a customer, I want to view my booking history so that I can review past reservations.

UC29	Submit Reviews	Customer	As a customer, I want to submit reviews so that I can share my experience.
UC30	Receive Notifications	Customer	As a customer, I want to receive notifications so that I can stay updated about bookings and services.
UC31	View Dashboard	Admin	As an administrator, I want to view the system dashboard so that I can monitor system activities.
UC32	View Account	Admin	As an administrator, I want to manage user accounts so that I can control system access.
UC33	Create Account	Admin	As an administrator, I want to create staff accounts so that employees can access the system.
UC34	Search Account	Admin	As an administrator, I want to search user accounts so that I can manage them efficiently.
UC35	Sort Account	Admin	As an administrator, I want to sort accounts so that I can find information quickly.
UC36	Block Account	Admin	As an administrator, I want to block accounts that violate system policies.
UC37	Backup Database	Admin	As an administrator, I want to back up the database so that system data is protected.
UC38	Restore Database	Admin	As an administrator, I want to restore the database so that data can be recovered when needed.
UC39	View Business Dashboard	Business Manager	As a business manager, I want to view the business dashboard so that I can monitor overall resort performance, including revenue statistics, occupancy rates, and operational indicators.
UC40	Manage Rooms	Business Manager	As a business manager, I want to manage room information so that room categories, pricing, availability, and operational data remain accurate and up to date.
UC41	Manage Tours	Business Manager	As a business manager, I want to manage tour packages so that tour information, schedules, pricing, and availability can be maintained effectively.
UC42	Manage Menu	Business Manager	As a business manager, I want to manage restaurant menu items so that food and beverage offerings remain accurate and updated.
UC43	Manage Tables	Business Manager	As a business manager, I want to manage restaurant tables so that seating arrangements and table availability can be organized efficiently.
UC44	View Financial Report	Business Manager	As a business manager, I want to view financial reports so that I can evaluate business performance, analyze revenue sources, and monitor financial results.
UC45	Export PDF/Excel Report	Business Manager	As a business manager, I want to export financial reports to PDF or Excel format so that business information can be archived, shared, and analyzed externally.
UC46	Manage Customer Reviews	Business Manager	As a business manager, I want to manage customer reviews so that guest feedback can be monitored and inappropriate content can be moderated.

UC47	View Customer Reviews	Business Manager	As a business manager, I want to view customer reviews so that I can understand customer satisfaction and identify areas for service improvement.
UC48	Approve/Hide Review	Business Manager	As a business manager, I want to approve or hide customer reviews so that only appropriate and relevant feedback is displayed on the public portal.
UC49	Manage Room Bookings	Receptionist	As a receptionist, I want to manage room bookings so that I can search, view, verify, and handle reservation information properly.
UC50	View Daily Arrival/Departure List	Receptionist	As a receptionist, I want to view daily arrivals and departures so that I can prepare front-desk operations for check-in and check-out.
UC51	Perform Customer Check-in	Receptionist	As a receptionist, I want to check guests in so that they can begin their stay after their booking and identity information are verified.
UC52	Submit Stay Registration	Receptionist	As a receptionist, I want to submit guest stay registration information so that the resort can comply with temporary residence regulations.
UC53	Perform Customer Check-out	Receptionist	As a receptionist, I want to check guests out so that their stay can be completed after all charges are reviewed and settled.
UC54	Generate Consolidated Invoice	Receptionist	As a receptionist, I want to generate a consolidated invoice including room charges, food orders, and tour bookings so that the guest can review the total amount payable.
UC55	Process Final Payment	Receptionist	As a receptionist, I want to process the final payment so that the consolidated invoice can be marked as paid and the check-out process can be completed.
UC56	Update Room Operational Status	Receptionist	As a receptionist, I want to update room operational statuses so that room availability and operational conditions remain accurate.
UC57	Create Maintenance Request	Receptionist	As a receptionist, I want to create maintenance requests when room issues are reported so that maintenance staff can inspect and resolve the problems.
UC58	Export Temporary Residence Report	Receptionist	As a receptionist, I want to export temporary residence reports so that guest stay information can be submitted to the required local authority when needed.
UC59	View Dashboard	Housekeeping Staff	As housekeeping staff, I want to view my dashboard so that I can monitor assigned tasks.
UC60	View Room Cleaning Tasks	Housekeeping Staff	As housekeeping staff, I want to view cleaning tasks so that I know which rooms require cleaning.
UC61	Update Room Cleaning Status	Housekeeping Staff	As housekeeping staff, I want to update cleaning status so that room availability is accurate.
UC62	View Cleaning	Housekeeping	As housekeeping staff, I want to view cleaning history so that I can track

	History	Staff	completed tasks.
UC63	Report Room Issues	Housekeeping Staff	As housekeeping staff, I want to report room issues so that maintenance staff can resolve them.
UC64	View Dashboard	F&B Staff	As an F&B Staff, I want to view the operational dashboard so that I can monitor food orders, table reservations, and restaurant activities in real time.
UC65	View Food Orders	F&B Staff	As an F&B Staff, I want to view all food orders so that I can track customer requests and manage the order workflow efficiently.
UC66	Update Food Order Status	F&B Staff	As an F&B Staff, I want to update the status of food orders (Pending → Cooking → Delivered) so that customers and other staff can track order progress.
UC67	View Table Reservations	F&B Staff	As an F&B Staff, I want to view table reservations so that I can prepare tables and manage restaurant seating arrangements.
UC68	Update Table Status	F&B Staff	As an F&B Staff, I want to update table status (Available, Reserved, Occupied) so that reservation information remains accurate.
UC69	Charge Food Bill To Room	F&B Staff	As an F&B Staff, I want to charge a customer's food bill directly to their room so that the amount can be included in the final consolidated invoice.
UC70	View Dashboard	Kitchen Staff	As a Kitchen Staff member, I want to view the kitchen dashboard so that I can monitor incoming orders and kitchen workload.
UC71	View Kitchen Order Tickets (KOT)	Kitchen Staff	As a Kitchen Staff member, I want to view all Kitchen Order Tickets (KOT) so that I can prepare food according to customer requests.
UC72	View KOT Details	Kitchen Staff	As a Kitchen Staff member, I want to view detailed information of a KOT so that I can prepare the correct dishes and quantities.
UC73	Update KOT Status	Kitchen Staff	As a Kitchen Staff member, I want to update the status of a KOT (Pending → Preparing → Ready to Serve) so that F&B Staff can know when orders are ready for delivery.

1.2.3 Use Case Diagrams

1.2.3.1 *UCs for Guest*

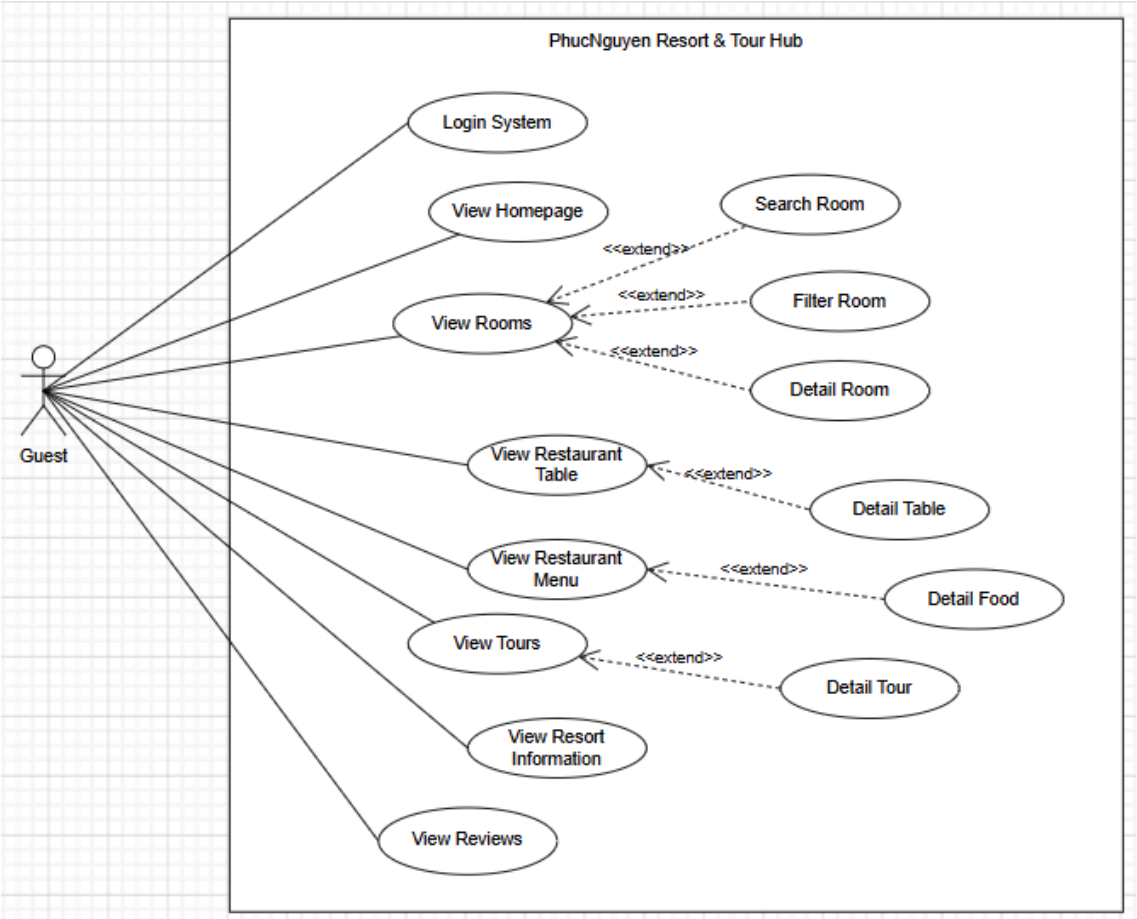


Figure I.2: Use Case Diagram - Guest

1.2.3.2 *UCs for Registered User*

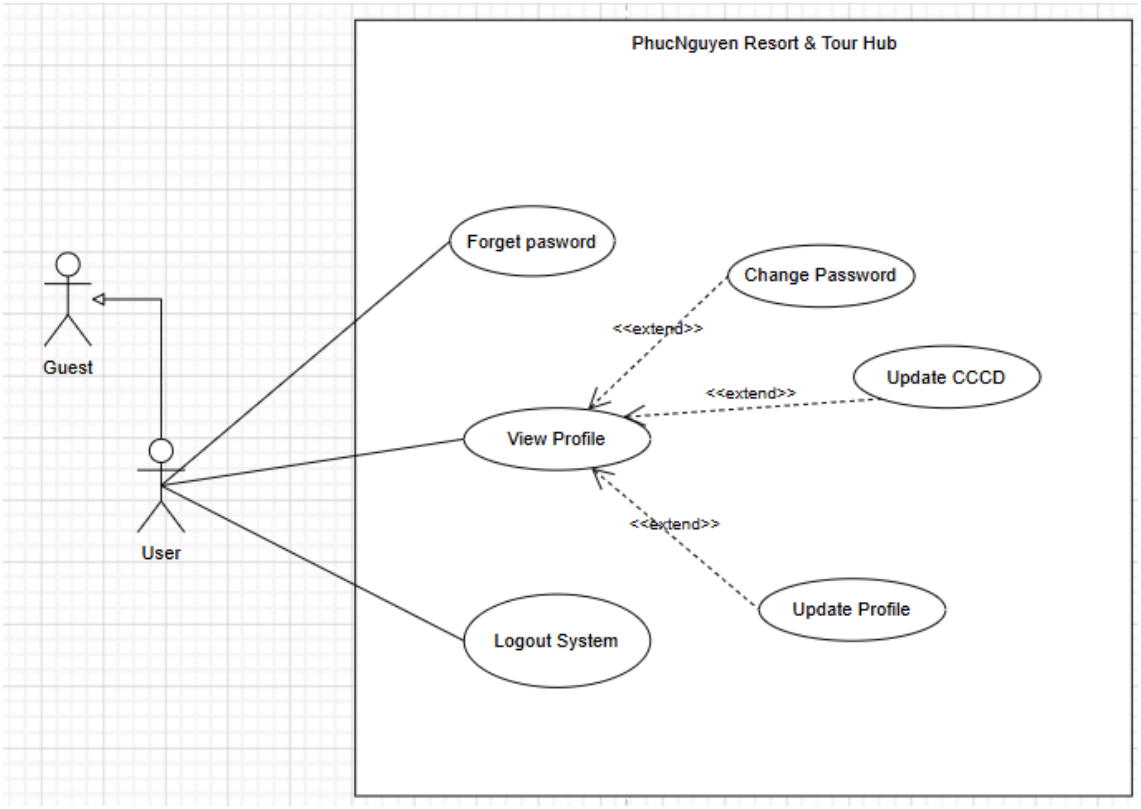


Figure I.3: Use Case Diagram - User

1.2.3.3 *UCs for Customer*

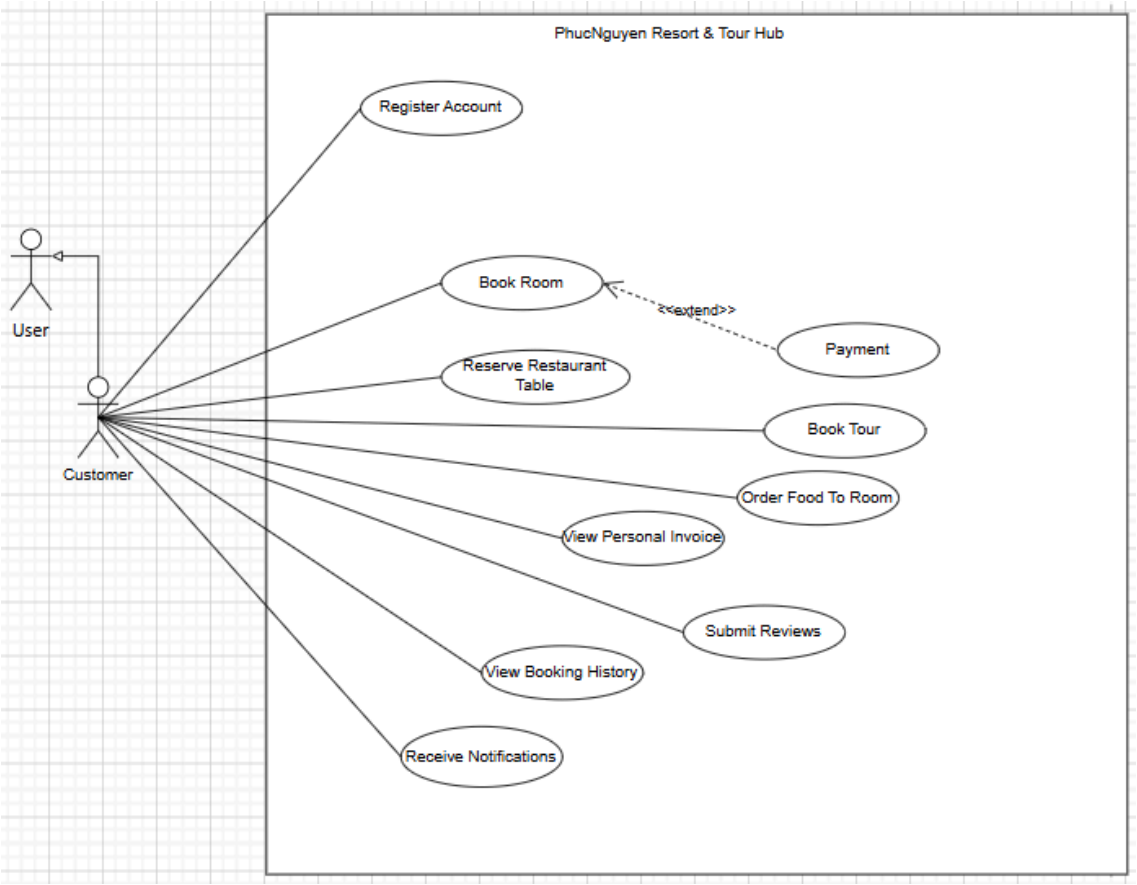


Figure I.4: Use Case Diagram - Customer

1.2.3.4 *UCs for Admin*

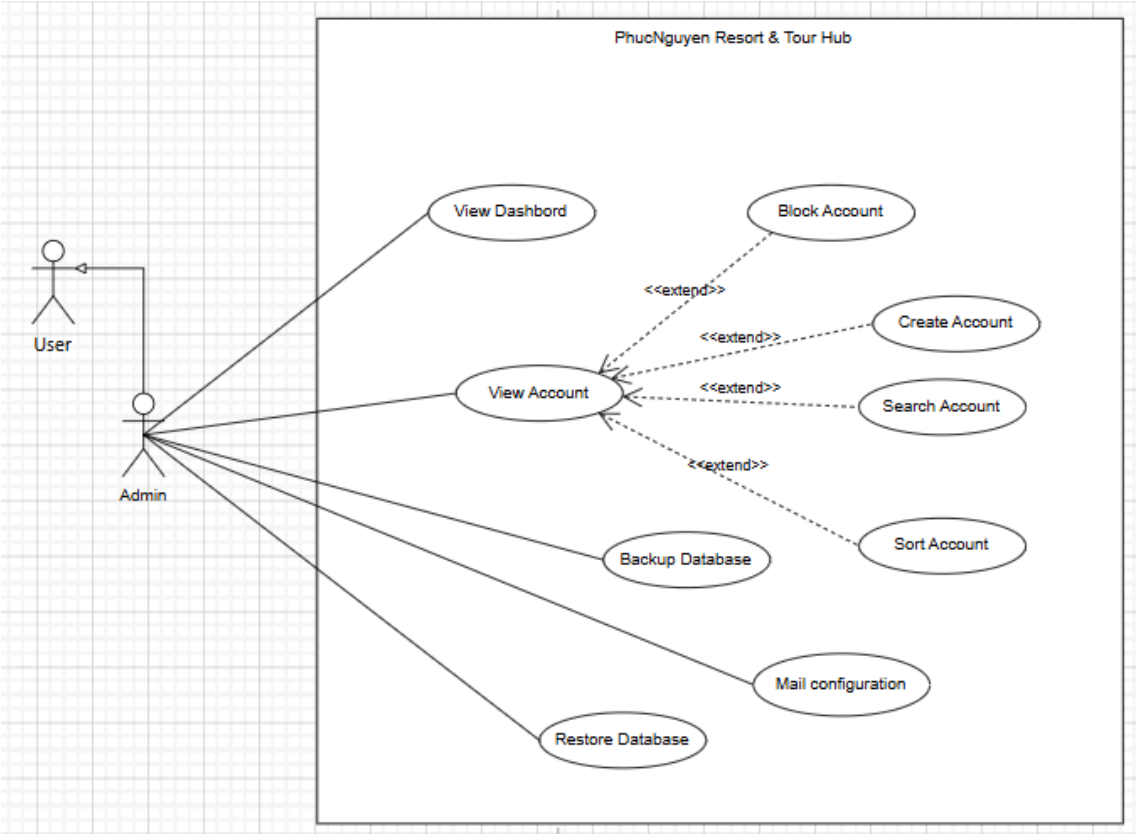


Figure I.5: Use Case Diagram - Admin

1.2.3.5 *UCs for Business Manager*

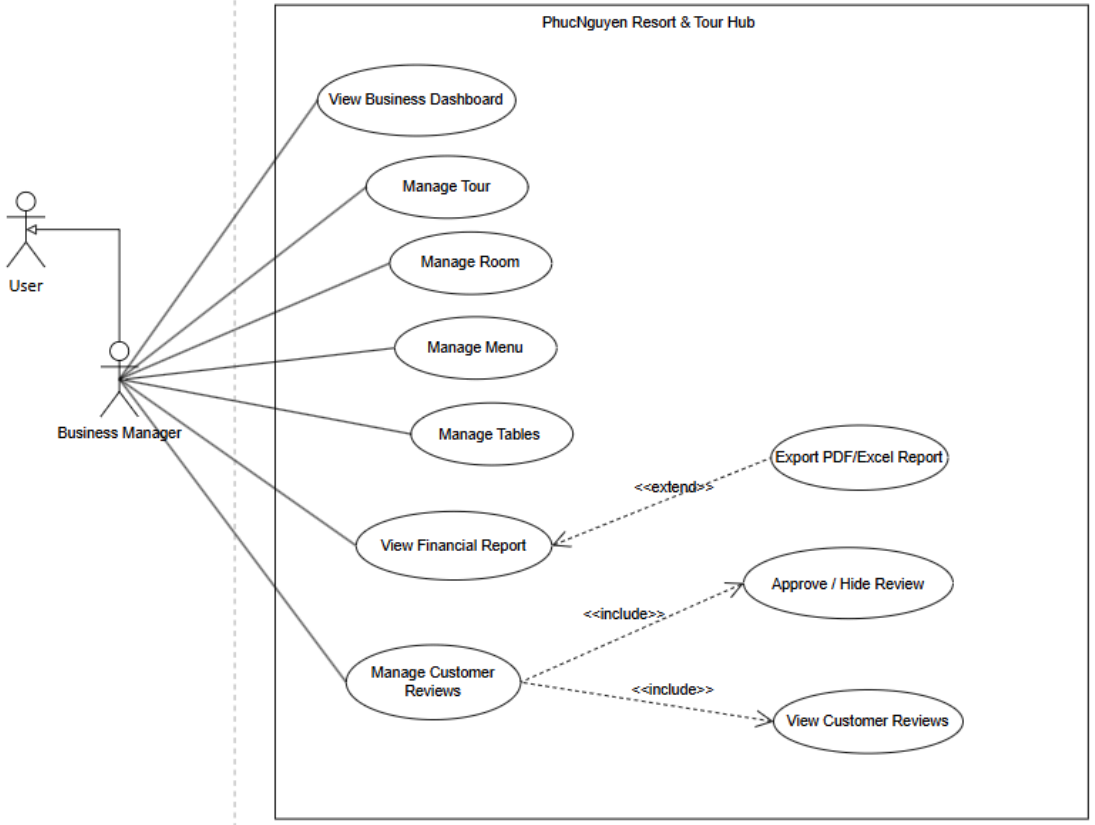


Figure I.6: Use Case Diagram -Business Manager

1.2.3.6 *UCs for Receptionist*

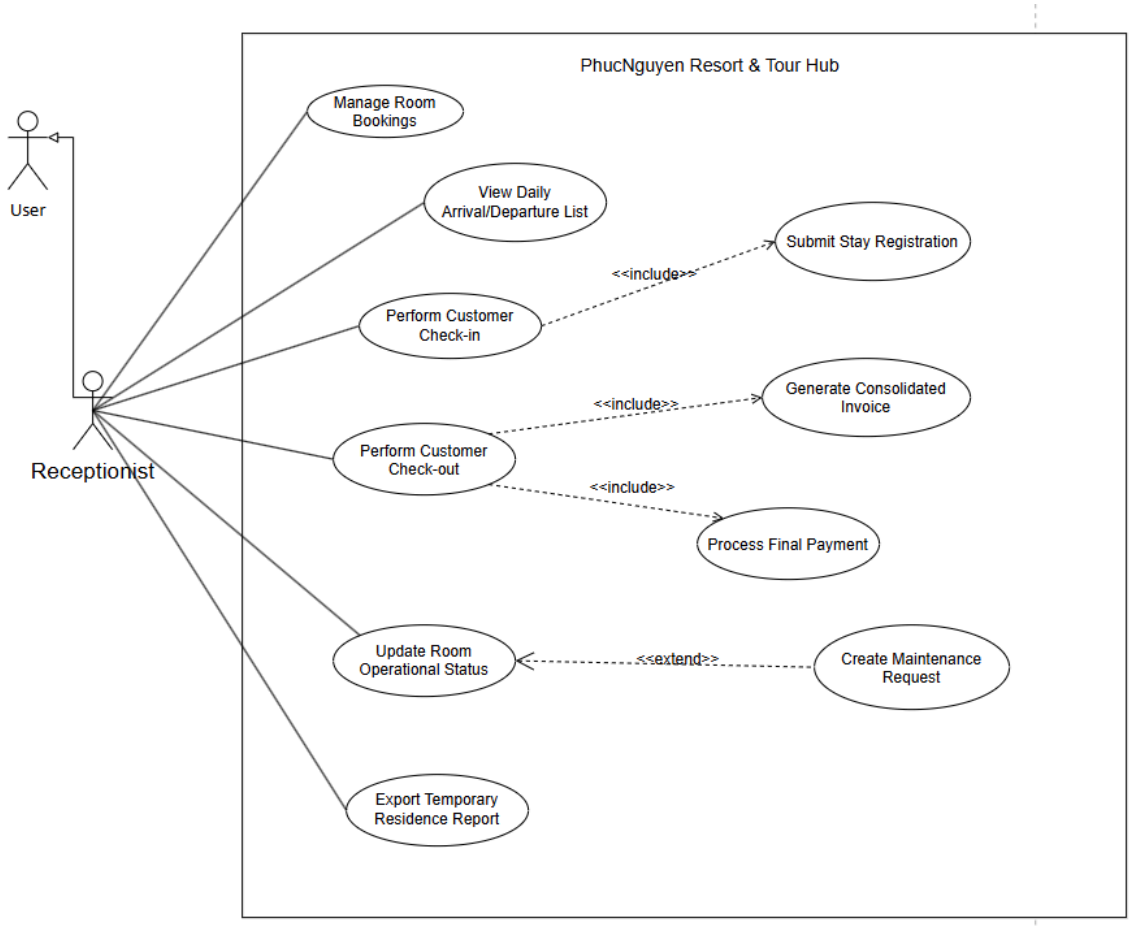


Figure I.7: Use Case Diagram - Receptionist

1.2.3.7 *UCs for Housekeeping Staff*

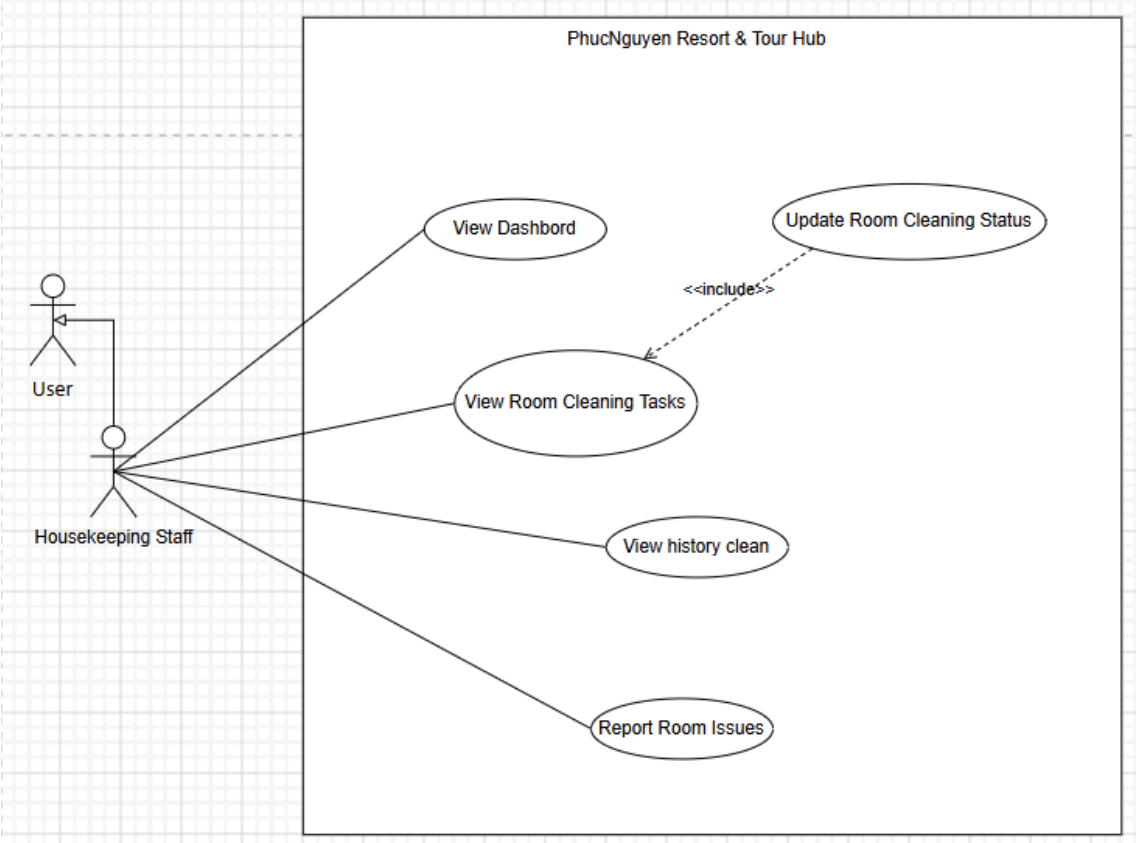


Figure I.8: Use Case Diagram - Housekeeping Staff

1.2.3.8 *UCs for Maintenance Staff*

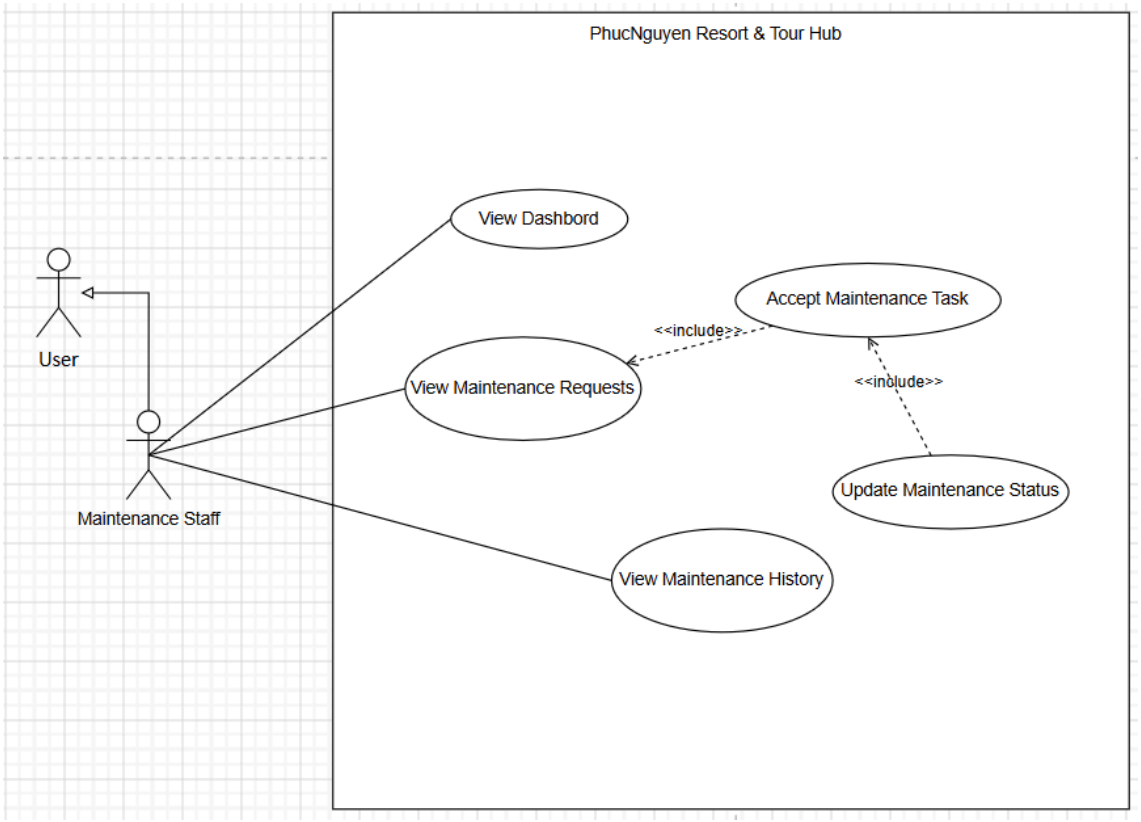


Figure I.9: Maintenance Staff

1.2.3.9 *UCs for F&B Staff*

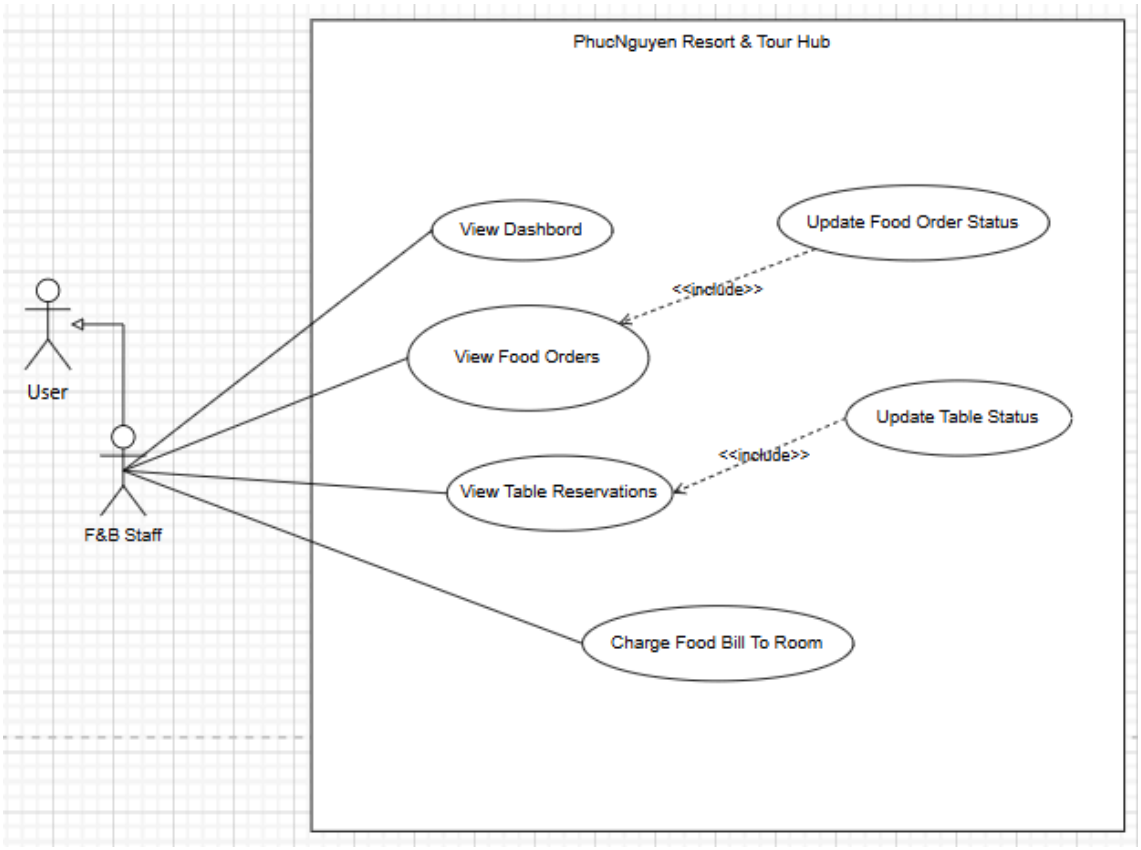


Figure I.10: F&B Staff

1.2.3.10 *UCs for Kitchen Staff*

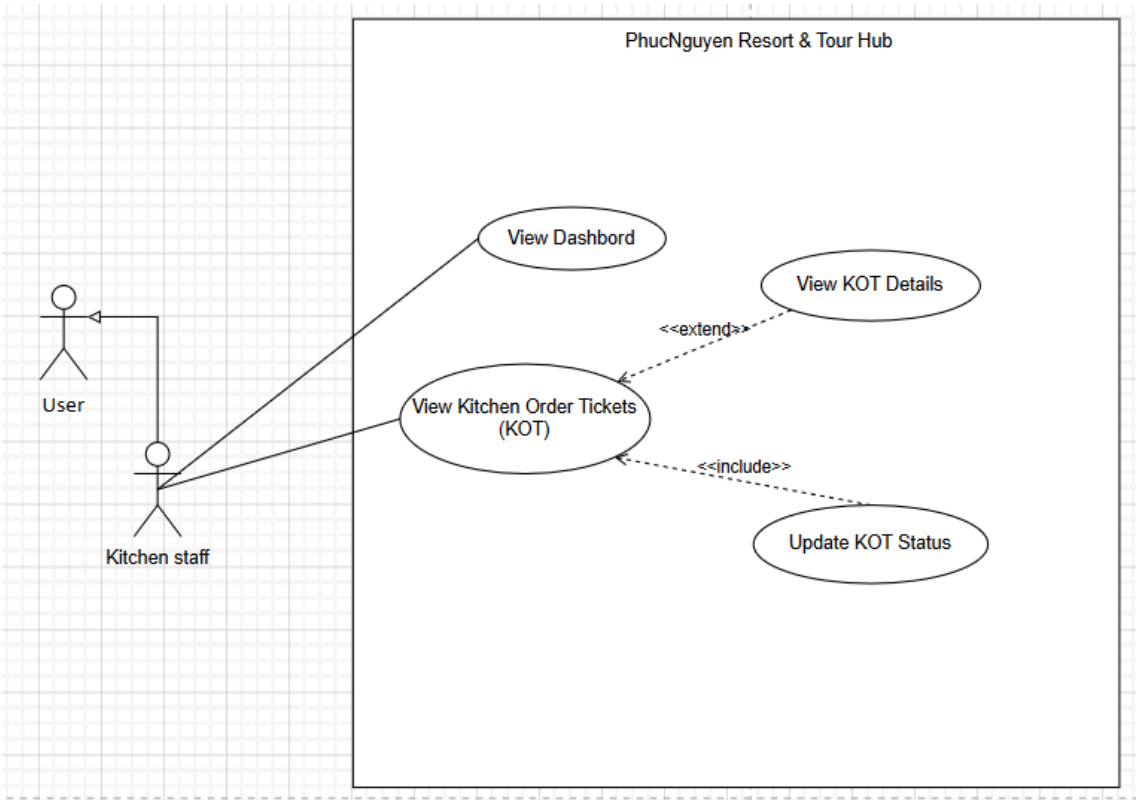
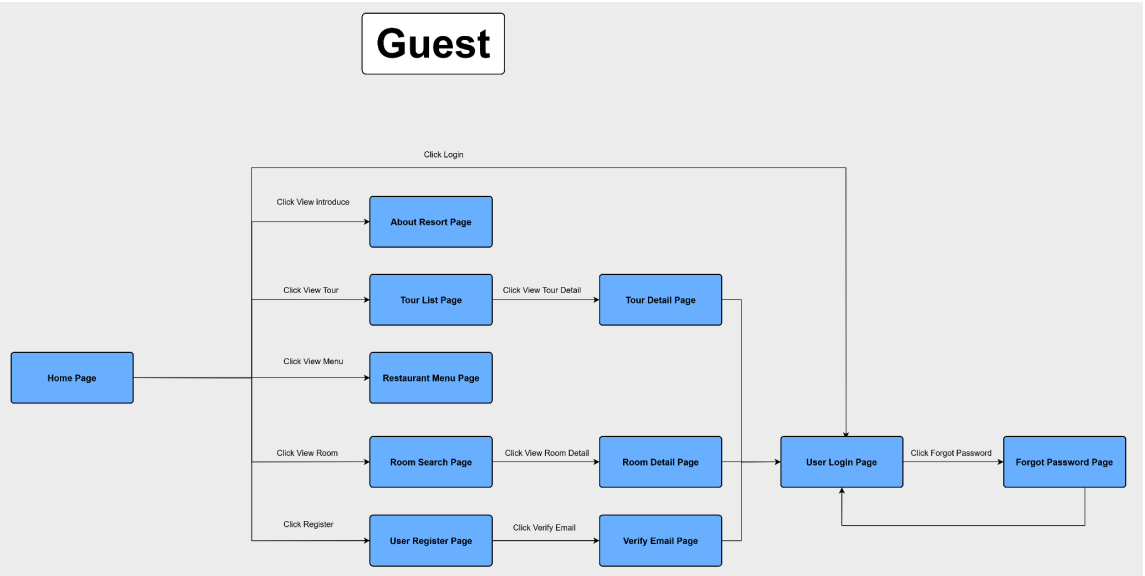


Figure I.11: Kitchen Staff

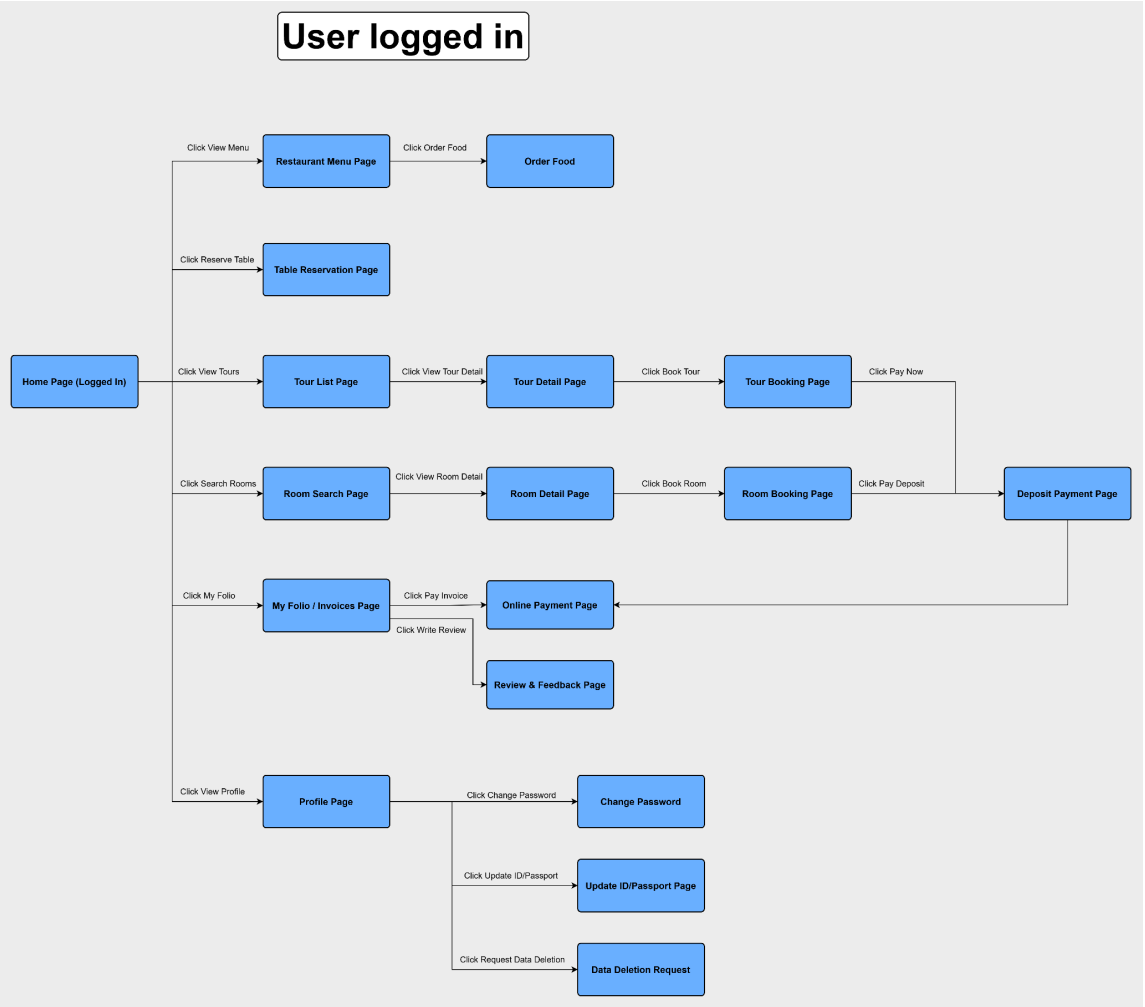
1.3 System Functionalities

1.3.1 Screens Flow

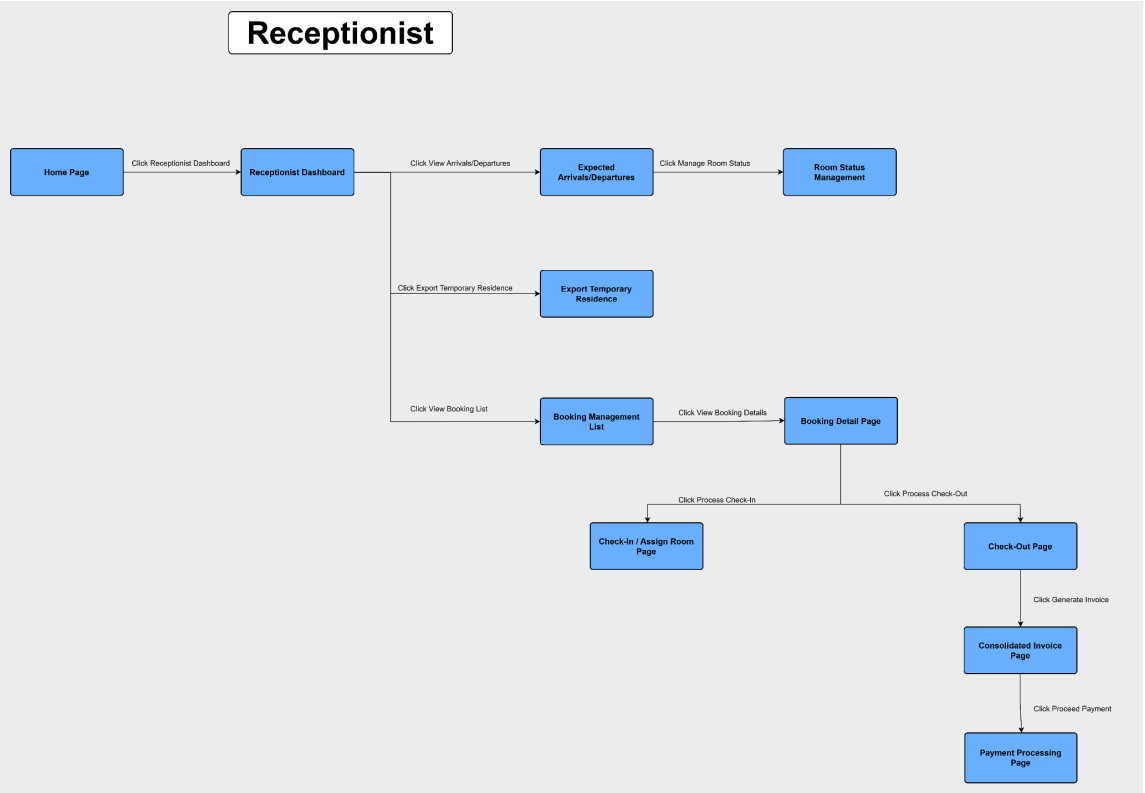
1.3.1.1 *Guest Screen Flow*



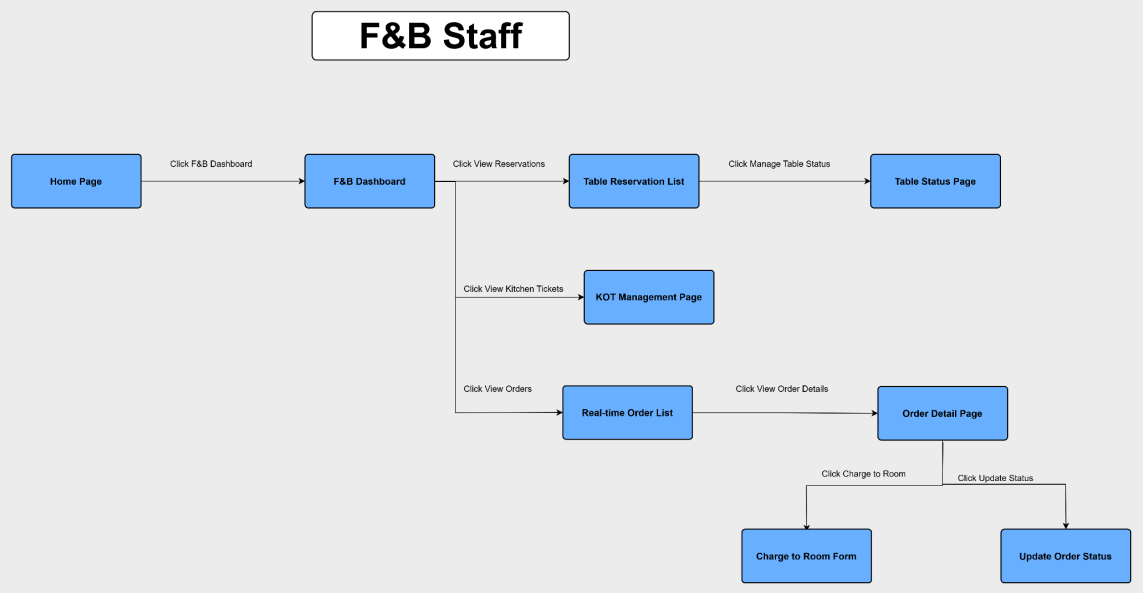
1.3.1.2 *User Logged In Screen Flow*



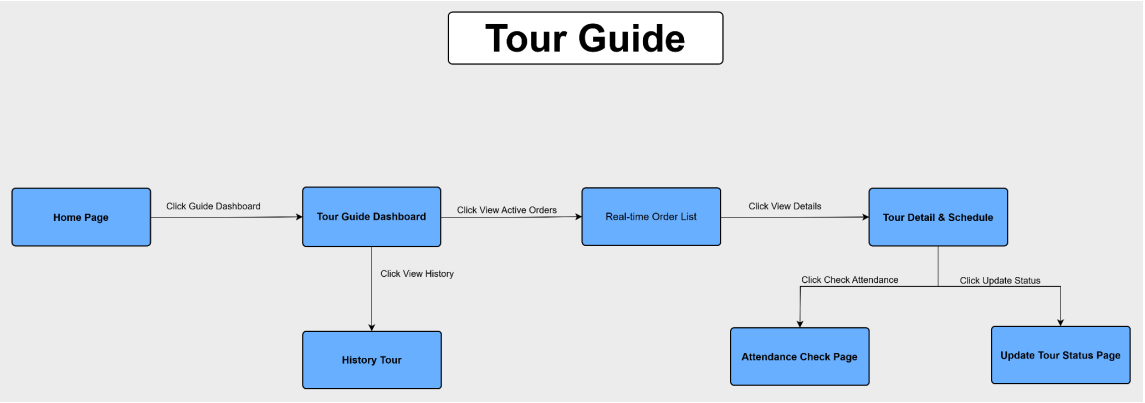
1.3.1.3 *Receptionist Screen Flow*



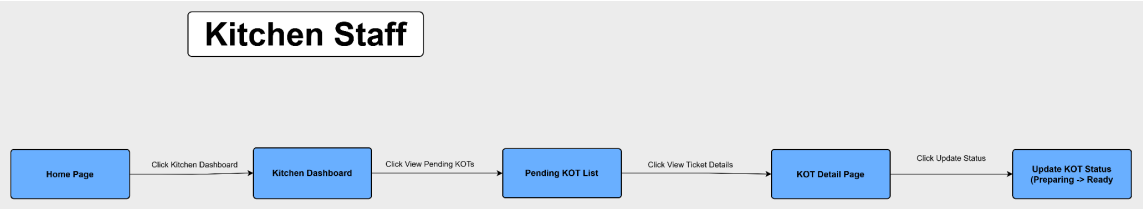
1.3.1.4 *F&B Staff Screen Flow*



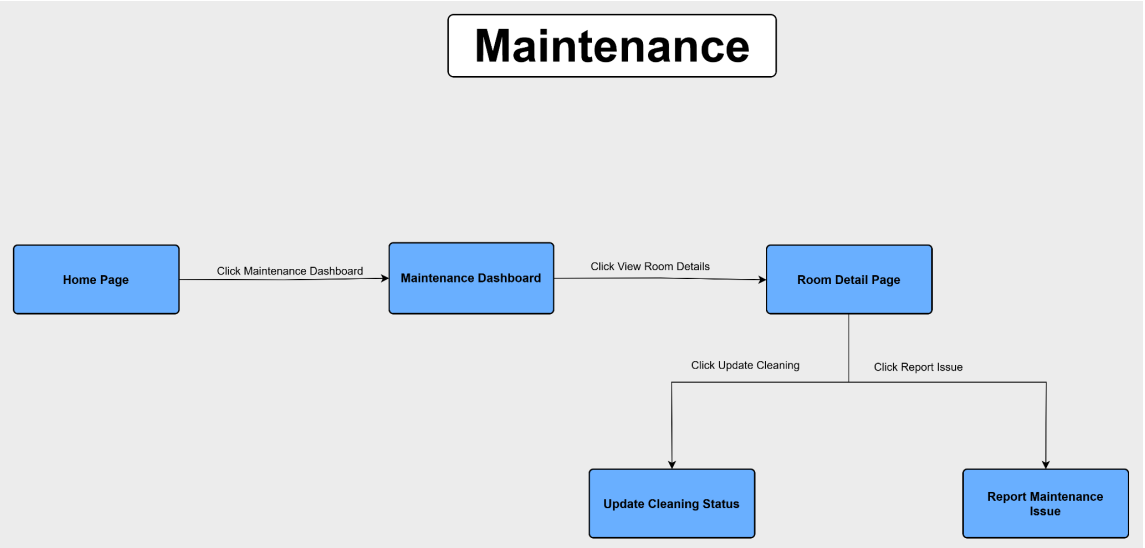
1.3.1.5 *Tour Guide Screen Flow*



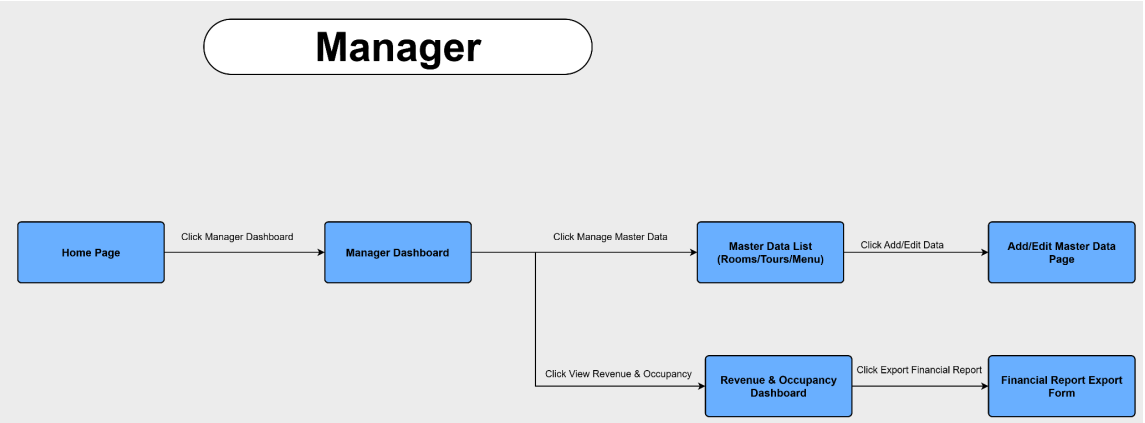
1.3.1.6 *Kitchen Staff Screen Flow*



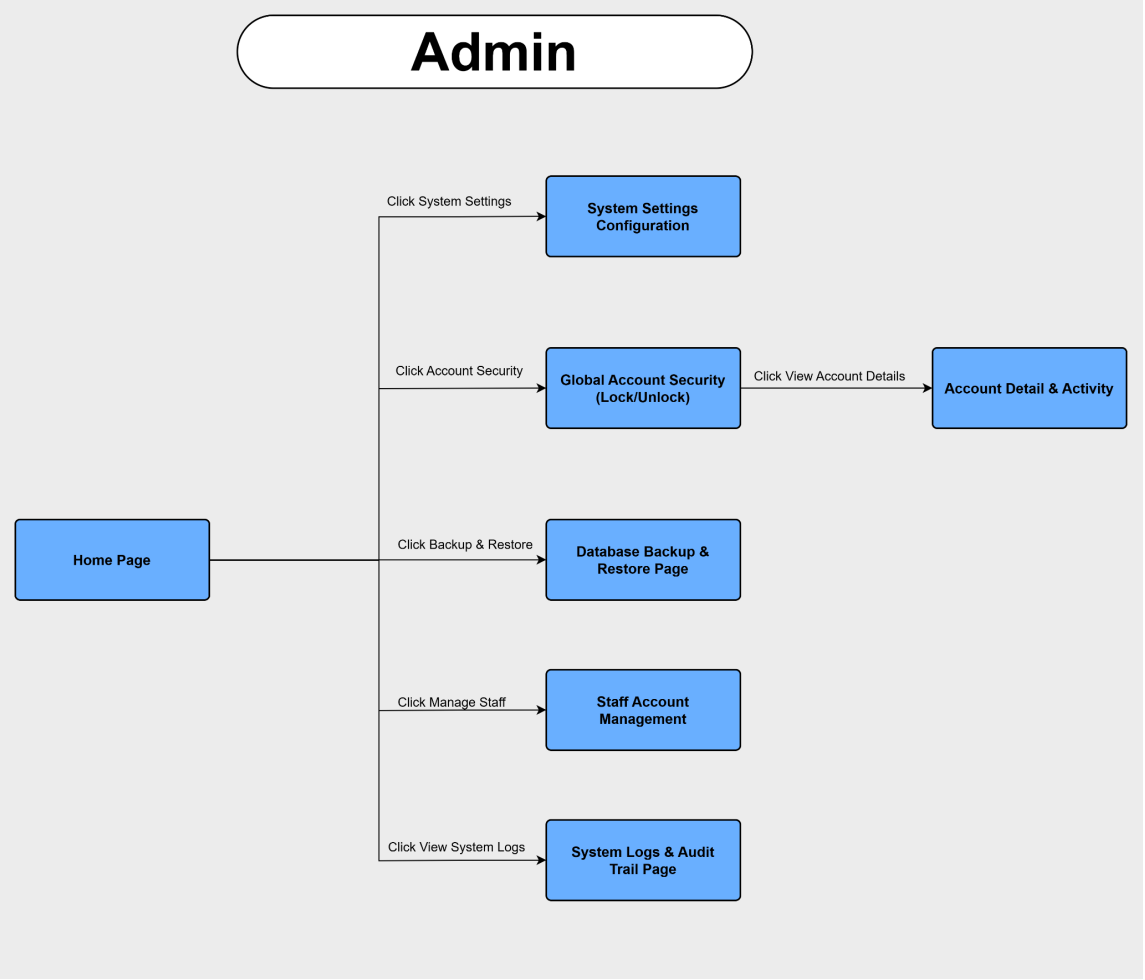
1.3.1.7 *Maintenance Screen Flow*



1.3.1.8 *Manager Screen Flow*



1.3.1.9 *Admin Screen Flow*



1.3.2 Screen Authorization

Screen Name	*Guest*	*Customer*	*Receptionist*	*F&B Staff*	*Kitchen Staff*	*Housekeeping*	*Maintenance*	*Tour Guide*	*Business Manager*	*Admin*
Home Page	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Login Page	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Register Account Page	✓									
Verify Email Page	✓	✓								
Forgot Password Page	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
About Resort Page	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Room Search Page	✓	✓								
Room Detail Page	✓	✓								
Tour List Page	✓	✓								
Tour Detail Page	✓	✓								
Restaurant Menu Page	✓	✓								
Food Detail Page	✓	✓								
Profile Page		✓	✓	✓	✓	✓	✓	✓	✓	✓

Change Password Page		✓	✓	✓	✓	✓	✓	✓	✓	✓
Update ID/Passport Page		✓								
Room Booking Page		✓	✓							
Deposit Payment Page		✓								
Table Reservation Page		✓		✓						
Order Food To Room Page		✓		✓						
Tour Booking Page		✓								
My Booking History Page		✓								
My Invoice Page		✓								
Receptionist Dashboard			✓							
Booking Management List			✓							
Check-in Page			✓							

Check-out Page			✓							
Consolidated Invoice Page			✓							
Payment Processing Page			✓							
F&B Dashboard				✓						
Food Order List				✓						
Food Order Detail				✓	✓					
Update Food Order Status				✓						
Kitchen Dashboard					✓					
Pending KOT List					✓					
KOT Detail Page				✓	✓					
Update KOT Status					✓					
Housekeeping Dashboard						✓				
Room Cleaning Task List						✓				
Update Cleaning Status						✓				

Maintenance Dashboard							✓			
Maintenance Request List			✓			✓	✓			
Tour Guide Dashboard								✓		
Assigned Tour List								✓		
Attendance Check								✓		
Business Manager Dashboard									✓	
Manage Rooms									✓	✓
Manage Tours									✓	✓
Manage Menu									✓	✓
Financial Reports									✓	✓
Review Moderation									✓	✓
Admin Dashboard										✓
Staff Account Management										✓
Database Backup & Restore										✓

System Logs & Audit Trail										✓

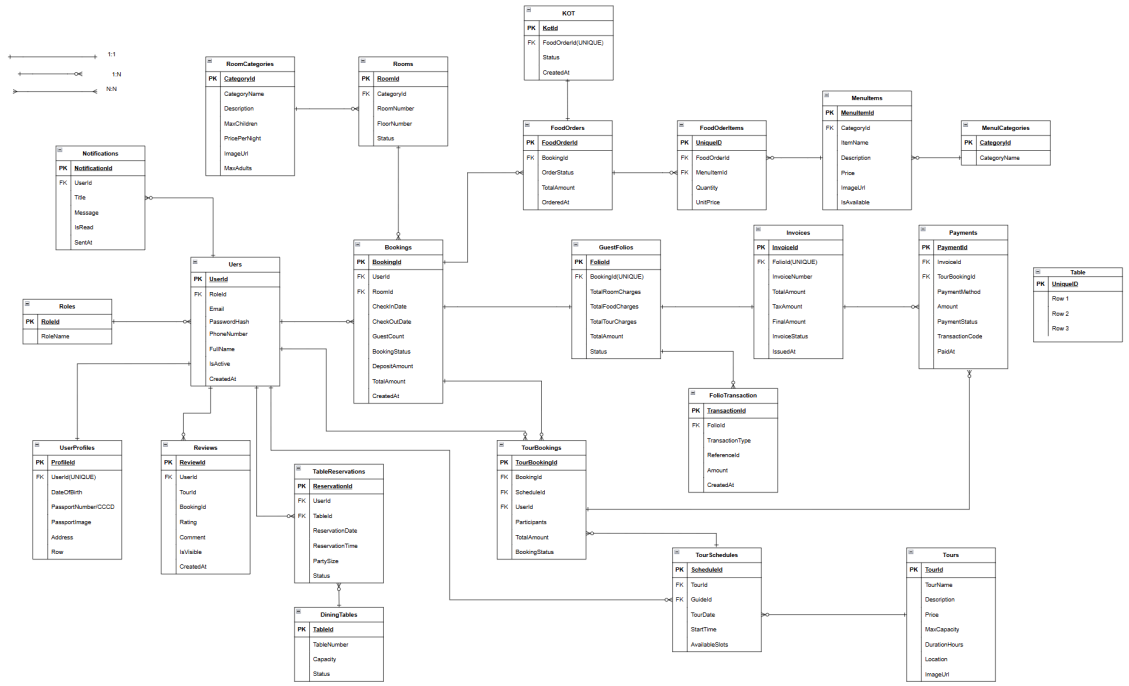
1.3.3 Non-UI Functions

#	Feature	System Function	Description
1	Authentication	Password Hashing	Hash passwords before storing.
2	Authentication	Email Verification Token	Generate verification tokens.
3	Authentication	Password Reset Token	Generate reset-password tokens.
4	Authorization	Role-Based Access Control	Authorize users by role.
5	Profile	Encrypt ID/Passport Data	Protect sensitive identity information.
6	Profile	Data Deletion Request	Process right-to-erasure requests.
7	Booking	Check Room Availability	Validate room availability.
8	Booking	Prevent Double Booking	Prevent overlapping reservations.
9	Booking	Booking Code Generation	Generate unique booking code.
10	Booking	Deposit Calculation	Calculate deposit amount.
11	Payment	Payment Gateway Integration	Connect VNPay/Stripe.
12	Payment	Payment Callback Handler	Handle payment response.
13	Notification	Booking Confirmation	Send email/SMS confirmation.
14	Notification	Digital Invoice Delivery	Send invoices electronically.
15	Front Desk	Assign Physical Room	Assign room during check-in.

16	Front Desk	Temporary Residence Validation	Validate residence registration data.
17	Room Status	State Transition Management	Manage room lifecycle status.
18	Housekeeping	Generate Cleaning Tasks	Create cleaning jobs automatically.
19	Maintenance	Maintenance Request Creation	Create maintenance tickets.
20	Maintenance	Maintenance Status Tracking	Track maintenance progress.
21	Restaurant	Table Availability Check	Check table availability.
22	Restaurant	Prevent Table Double Booking	Prevent table overlaps.
23	Restaurant	Food Order Creation	Create food orders.
24	Kitchen	Generate KOT	Generate Kitchen Order Tickets.
25	Kitchen	KOT Status Synchronization	Update KOT workflow status.
26	F&B	Charge To Room	Push restaurant charges to room folio.
27	Tour	Tour Schedule Validation	Validate tour capacity and schedule.
28	Tour	Weather API Integration	Retrieve weather information.
29	Tour	Map API Integration	Display map/location information.
30	Feedback	Review Moderation Logic	Control review visibility.
31	Billing	Guest Folio Creation	Create folio per stay.
32	Billing	Add Room Charges	Post room charges.
33	Billing	Add Food Charges	Post F&B charges.
34	Billing	Add Tour Charges	Post tour charges.

35	Billing	Outstanding Charge Validation	Prevent checkout with unpaid items.
36	Billing	Consolidated Invoice Generation	Generate final invoice.
37	Billing	Final Payment Processing	Process checkout payment.
38	Refund	Refund Calculation	Apply cancellation policy.
39	Dashboard	Occupancy Rate Calculation	Calculate occupancy KPIs.
40	Dashboard	Revenue Breakdown Calculation	Calculate revenue by source.
41	Report	Excel/PDF Export	Export reports.
42	Admin	Staff Account Creation	Create employee accounts.
43	Admin	Account Lock/Unlock	Manage account security.
44	Admin	Database Backup	Backup database.
45	Admin	Database Restore	Restore database.
46	Audit	System Activity Logging	Record user activities.
47	Audit	Error Logging	Record application errors.
48	Night Audit	Daily Closing Process	Finalize daily business data.
49	Notification	Tour Ticket QR Generation	Generate QR tickets.
50	Security	Session Management	Manage login sessions.

1.4 Entity Relationship Diagram



Entity Relationship Diagram

Entities Description

No	Table	Description
1	Roles	• Primary keys: RoleId • Foreign keys: None • Unique keys: RoleName
2	Users	• Primary keys: UserId • Foreign keys: RoleId → Roles(RoleId) • Unique keys: Email
3	UserProfiles	• Primary keys: ProfileId • Foreign keys: UserId → Users(UserId) • Unique keys: UserId
4	RoomCategories	• Primary keys: CategoryId • Foreign keys: None • Unique keys: None
5	Rooms	• Primary keys: RoomId • Foreign keys: CategoryId → RoomCategories(CategoryId) • Unique keys: RoomNumber
6	Bookings	• Primary keys: BookingId • Foreign keys: UserId → Users(UserId), RoomId → Rooms(RoomId) • Unique keys: None

7	BookingGuests	• Primary keys: BookingGuestId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: None
8	MenuCategories	• Primary keys: CategoryId • Foreign keys: None • Unique keys: None
9	MenuItems	• Primary keys: MenuItemId • Foreign keys: CategoryId → MenuCategories(CategoryId) • Unique keys: None
10	DiningTables	• Primary keys: TableId • Foreign keys: None • Unique keys: None
11	TableReservations	• Primary keys: ReservationId • Foreign keys: UserId → Users(UserId), TableId → DiningTables(TableId) • Unique keys: None
12	FoodOrders	• Primary keys: FoodOrderId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: None
13	FoodOrderItems	• Primary keys: OrderItemId • Foreign keys: FoodOrderId → FoodOrders(FoodOrderId), MenuItemId → MenuItems(MenuItemId) • Unique keys: None
14	KOT	• Primary keys: KotId • Foreign keys: FoodOrderId → FoodOrders(FoodOrderId) • Unique keys: FoodOrderId
15	Tours	• Primary keys: TourId • Foreign keys: None • Unique keys: None
16	TourSchedules	• Primary keys: ScheduleId • Foreign keys: TourId → Tours(TourId), GuideId → Users(UserId) • Unique keys: None
17	TourBookings	• Primary keys: TourBookingId • Foreign keys: UserId → Users(UserId), BookingId (Nullable) → Bookings(BookingId), ScheduleId → TourSchedules(ScheduleId) • Unique keys: None
18	Reviews	• Primary keys: ReviewId • Foreign keys: UserId → Users(UserId) • Unique keys: None
19	GuestFolios	• Primary keys: FolioId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: BookingId
20	FolioTransactions	• Primary keys: TransactionId • Foreign keys: FolioId → GuestFolios(FolioId)

		• Unique keys: None
21	Invoices	• Primary keys: InvoiceId • Foreign keys: FolioId → GuestFolios(FolioId) • Unique keys: FolioId
22	Payments	• Primary keys: PaymentId • Foreign keys: InvoiceId (Nullable) → Invoices(InvoiceId), TourBookingId (Nullable) → TourBookings(TourBookingId) • Unique keys: None
23	Refunds	• Primary keys: RefundId • Foreign keys: BookingId → Bookings(BookingId) • Unique keys: None
24	Notifications	• Primary keys: NotificationId • Foreign keys: UserId → Users(UserId) • Unique keys: None

2. Use Case Specifications

2.1 Auth, Profiles & Master Data

2.1.1 UC21 – Register Account

Field	Specification
Use Case ID	UC21
Use Case Name	Register Account
Feature / Module	Auth, Profiles & Master Data
Primary Actor	Guest
Secondary Actors	Email Verification Service
Description	As a guest, I want to create an account so that I can use resort services such as booking rooms, reserving tables, ordering food, booking tours, and viewing invoices.

Preconditions	1. Guest is not logged in. 2. Guest accesses the Register page. 3. The email entered by the guest has not been used by another account. 4. The system is connected to the email verification service.
Postconditions	1. A new user account is created with Customer role. 2. Account status is set to Pending Verification or Inactive until email verification is completed. 3. A verification email is sent to the registered email address. 4. Registration activity is recorded in the system log.
Normal Sequence / Flow	1. Guest clicks Register from the page header or registration entry point. 2. System displays the Register Account screen. 3. Guest enters registration information, including full name, email, phone number, password, and confirm password. 4. Guest clicks the Register button. 5. System validates required fields. 6. System validates email format. 7. System validates phone number format. 8. System validates password strength. 9. System checks whether password and confirm password match. 10. System checks whether the email already exists in the database. 11. System hashes the password before storing it. 12. System creates a new user account with role Customer. 13. System creates the corresponding user profile record. 14. System generates an email verification token.

	15. System sends a verification email to the guest. 16. System displays a successful registration message and asks the guest to verify email before logging in.
Alternative Sequence / Flow A1 – Email already exists	At step 10, if the email already exists: 1. System does not create a new account. 2. System displays message: “This email is already registered.” 3. Guest may return to the login page or use forgot password.
Alternative Sequence / Flow A2 – Invalid input data	At steps 5–9, if input data is invalid: 1. System highlights invalid fields. 2. System displays the corresponding validation message. 3. Guest corrects the information. 4. Flow returns to step 4 of the normal flow.
Alternative Sequence / Flow A3 – Email verification service unavailable	At step 15, if the email service is unavailable: 1. System creates the account with Pending Verification status. 2. System records the failed email sending event. 3. System displays message: “Account created, but verification email could not be sent. Please request email verification again later.”
Exception Flow E1 - System/DB error	If a system or database error occurs during account creation: 1. System rolls back account creation transaction. 2. System displays a general error message. 3. System records the error in the system log.
Business Rules	BR-UC21-01: Email must be unique in the system. BR-UC21-02: Password must satisfy the system password policy. BR-UC21-03: Password must be stored using hashing, not plain text.

	BR-UC21-04: New registered account must not access authenticated customer features until email verification is completed. BR-UC21-05: Default role for self-registration is Customer.
Related Use Cases	UC01 Login UC19 Forgot Password UC20 Logout UC16 Update Profile UC17 Update CCCD/Passport
Related Components	**Controllers**: AuthController, RegisterController **Services:** AuthService, UserService, EmailVerificationService, PasswordHashingService **Repositories / DAO:** UserRepository, UserProfileRepository, RoleRepository, VerificationTokenRepository **Entities / Tables:** Users, UserProfiles, Roles, VerificationTokens, ActivityLogs
Input Data	Full name, email, phone number, password, confirm password
Output Data	Registration result message, verification email, pending customer account
Priority	High
Frequency of Use	Medium

2.1.2 UC01 – Login

Field	Specification
Use Case ID	UC01

Use Case Name	Login
Feature / Module	Auth, Profiles & Master Data
Primary Actor	Registered User
Secondary Actors	None
Description	As a registered user, I want to log into the system so that I can access system features according to my assigned role.
Preconditions	1. User account already exists. 2. User account is active. 3. User is not currently logged in.
Postconditions	1. User logs in successfully. 2. Authenticated session is created. 3. User role and permissions are loaded. 4. Login activity is recorded.
Input Data	Email, password
Output Data	Login result message, authenticated session, user profile, user role, access permissions
Normal Sequence / Flow	1. User opens the Login page. 2. System displays the Login form. 3. User enters email and password. 4. User submits the login request. 5. System authenticates the user information. 6. System creates an authenticated session. 7. System redirects the user to the appropriate page based on role.
Alternative Sequence / Flow A1 – Invalid login information	At step 5, if email or password is incorrect: 1. System rejects the login request. 2. System displays an invalid login message. 3. User may enter login information again.
Alternative Sequence / Flow A2 – Account inactive or blocked	At step 5, if the account is inactive or blocked: 1. System denies access. 2. System displays an account status warning message.
Alternative Sequence / Flow A3 – Email not verified	At step 5, if the email has not been verified: 1. System denies access. 2. System asks the user to verify email before logging in.

Exception Flow E1 – System error	If system error occurs during login: 1. System does not create a session. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC01-01: Email and password are required. BR-UC01-02: Password must be verified against stored hashed password. BR-UC01-03: Only active accounts can access the system. BR-UC01-04: User permissions are determined by assigned role. BR-UC01-05: Failed login attempts should be tracked for security.
Related Use Cases	UC21 Register Account UC19 Forgot Password UC20 Logout
Related Components	Controllers: AuthController, LoginController Services: AuthService, UserService, RoleService, PasswordHashingService Repositories / DAO: UserRepository, RoleRepository, ActivityLogRepository Entities / Tables: Users, Roles, ActivityLogs
Priority	High
Frequency of Use	Very High
Entities Affected	Users, Roles, ActivityLogs

2.1.3 UC17 – Update CCCD/Passport

Field	Specification
Use Case ID	UC17
Use Case Name	Update CCCD/Passport
Feature / Module	Auth, Profiles & Master Data
Primary Actor	Registered User
Secondary Actors	File Storage Service

Description	As a registered user, I want to upload or update my CCCD/Passport information so that my identity can be verified for booking, check-in, and stay registration processes.
Preconditions	1. User has logged into the system. 2. User profile already exists. 3. User accesses the Update CCCD/Passport page. 4. User agrees to provide identity information for verification purposes.
Postconditions	1. CCCD/Passport information is saved or updated. 2. Uploaded document file is stored securely. 3. Identity verification status is updated. 4. Update activity is recorded.
Input Data	Document type, CCCD/Passport number, full name on document, nationality, date of birth, issue date, expiry date, uploaded document image/file, consent confirmation
Output Data	Updated identity document record, updated verification status, success/error message
Normal Sequence / Flow	1. User opens the Profile page. 2. User selects Update CCCD/Passport. 3. System displays the identity document form. 4. User enters document information and uploads document file. 5. User confirms consent and submits the form. 6. System validates the submitted information and file. 7. System saves or updates the identity document information securely. 8. System displays the update result.
Alternative Sequence / Flow A1 – Invalid document information	At step 6, if document information is invalid: 1. System rejects the update. 2. System displays validation messages. 3. User corrects the information and submits again.
Alternative Sequence / Flow A2 – Invalid file upload	At step 6, if uploaded file is invalid: 1. System rejects the file. 2. System displays an invalid file message. 3. User uploads another file.

Alternative Sequence / Flow A3 – Duplicate document number	At step 6, if the document number is already used by another account: 1. System does not save the update. 2. System displays duplicate document message.
Exception Flow E1 – File storage error	If the document file cannot be stored: 1. System cancels the update. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC17-01: User must provide consent before identity document information is stored. BR-UC17-02: CCCD/Passport number must be unique among active accounts. BR-UC17-03: Only allowed file types can be uploaded. BR-UC17-04: Sensitive identity data must be stored securely. BR-UC17-05: Updating identity information changes verification status to pending until verified.
Related Use Cases	UC16 Update Profile UC51 Perform Customer Check-in UC52 Submit Stay Registration
Related Components	Controllers: ProfileController, IdentityDocumentController Services: UserProfileService, IdentityDocumentService, FileStorageService, EncryptionService Repositories / DAO: UserProfileRepository, IdentityDocumentRepository, UserRepository, ActivityLogRepository Entities / Tables: Users, UserProfiles, IdentityDocuments, ActivityLogs
Priority	High
Frequency of Use	Low
Entities Affected	Users, UserProfiles, IdentityDocuments, ActivityLogs

2.1.4 UC33 – Create Staff Account

Field	Specification
Use Case ID	UC33
Use Case Name	Create Staff Account
Feature / Module	Auth, Profiles & Master Data

Primary Actor	Admin
Secondary Actors	Email Service
Description	As an administrator, I want to create staff accounts so that resort employees can access the system according to their assigned roles and responsibilities.
Preconditions	1. Admin has logged into the system. 2. Admin has permission to manage staff accounts. 3. Staff role already exists in the system.
Postconditions	1. New staff account is created. 2. Staff role is assigned. 3. Staff account status is set according to system policy. 4. Activation or account information email is sent to the staff member. 5. Account creation activity is recorded.
Input Data	Full name, email, phone number, staff role, initial status, optional initial password
Output Data	Created staff account, assigned role, activation email, success/error message
Normal Sequence / Flow	1. Admin opens Staff Account Management page. 2. System displays staff account management screen. 3. Admin selects Create Staff Account. 4. Admin enters staff information and selects staff role. 5. Admin submits the account creation request. 6. System validates staff information and role. 7. System creates the staff account and assigns the selected role. 8. System sends activation or account information email. 9. System displays account creation result.
Alternative Sequence / Flow A1 – Email already exists	At step 6, if the email already exists: 1. System rejects account creation. 2. System displays duplicate email message. 3. Admin may correct the email and submit again.
Alternative Sequence / Flow A2 – Invalid role selected	At step 6, if the selected role is invalid or inactive: 1. System rejects account creation. 2. System displays invalid role message. 3. Admin selects another role.

Alternative Sequence / Flow A3 – Email service unavailable	At step 8, if email sending fails: 1. System keeps the account created with pending activation status. 2. System records email sending failure. 3. System displays a warning message.
Exception Flow E1 – Database error	If a database error occurs during account creation: 1. System rolls back account creation. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC33-01: Only Admin can create staff accounts. BR-UC33-02: Staff email must be unique. BR-UC33-03: Staff account must be assigned a valid role. BR-UC33-04: Password must be hashed before storage. BR-UC33-05: Staff may be required to change password on first login.
Related Use Cases	UC01 Login UC32 View Account UC34 Search Account UC36 Block Account
Related Components	Controllers: AdminController, StaffAccountController Services: UserService, StaffAccountService, RoleService, EmailService, PasswordHashingService Repositories / DAO: UserRepository, RoleRepository, UserProfileRepository, ActivityLogRepository Entities / Tables: Users, Roles, UserProfiles, ActivityLogs
Priority	High
Frequency of Use	Medium
Entities Affected	Users, Roles, UserProfiles, ActivityLogs

2.1.5 UC40 – Manage Rooms

Field	Specification
Use Case ID	UC40
Use Case Name	Manage Rooms
Feature / Module	Auth, Profiles & Master Data
Primary Actor	Business Manager

Secondary Actors	File Storage Service
Description	As a business manager, I want to manage room master data so that room information, room categories, pricing, capacity, images, and availability status remain accurate for room reservation and operational management.
Preconditions	1. Business Manager is logged into the system. 2. Business Manager has permission to manage room data. 3. Room category information already exists in the system.
Postconditions	1. Room information is created, updated, enabled, disabled, or viewed successfully. 2. Room catalog is updated and available for related modules. 3. Room management activity is recorded in the system log.
Input Data	Room number, room category, room description, room capacity, price per night, room images, room status
Output Data	Updated room record, updated room catalog, success/error message
Normal Sequence / Flow	1. Business Manager opens Room Management. 2. System displays room information. 3. Business Manager selects an operation (Create, View, Update, Enable, Disable) .4. Business Manager enters or updates room information. 5. Business Manager submits the changes. 6. System validates the room information. 7. System saves the changes. 8. System displays the operation result.
Alternative Sequence / Flow A1 – Invalid room information	At step 6, if room information is invalid: 1. System rejects the request. 2. System displays validation messages. 3. Business Manager updates the information and submits again.
Alternative Sequence / Flow A2 – Duplicate room number	At step 6, if room number already exists: 1. System rejects the request. 2. System displays duplicate room number message. 3. Business Manager provides another room number.
Alternative Sequence / Flow A3 – Room linked to active bookings	If Business Manager attempts to disable or remove a room with active or future bookings: 1. System checks related bookings. 2. System prevents the operation according to business policy. 3. System displays warning message.

Exception Flow E1 – System error	If a system or database error occurs: 1. System cancels the operation. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC40-01: Room number must be unique. BR-UC40-02: Room capacity must be greater than zero. BR-UC40-03: Room price must be greater than zero. BR-UC40-04: Rooms with historical booking records must not be physically deleted. BR-UC40-05: Only active rooms can be displayed in room search and booking functions.
Related Use Cases	UC04 Search Available RoomsUC22 Book RoomUC51 Perform Customer Check-inUC56 Update Room Operational Status
Related Components	Controllers: RoomController, RoomManagementController Services: RoomService, RoomCategoryService, FileStorageService Repositories / DAO: RoomRepository, RoomCategoryRepository, BookingRepository, ActivityLogRepository Entities / Tables: Rooms, RoomCategories, Bookings, ActivityLogs
Priority	High
Frequency of Use	Medium
Entities Affected	Rooms, RoomCategories, Bookings, ActivityLogs

2.1.6 UC41 – Manage Tours

Field	Specification
Use Case ID	UC41
Use Case Name	Manage Tours
Feature / Module	Auth, Profiles & Master Data
Primary Actor	Business Manager
Secondary Actors	File Storage Service

Description	As a business manager, I want to manage tour master data so that tour information, schedules, capacities, pricing, and availability remain accurate for customers.
Preconditions	1. Business Manager is logged into the system.2. Business Manager has permission to manage tour data.
Postconditions	1. Tour information is created, updated, enabled, disabled, or viewed successfully. 2. Tour catalog is updated.
Input Data	Tour name, description, itinerary, duration, price, capacity, schedule, image, status
Output Data	Updated tour record, updated tour catalog, success/error message
Normal Sequence / Flow	Similar to UC40 – Manage Rooms. The managed entity is **Tour** instead of **Room**.
Alternative Sequence / Flow	Similar to UC40 – Manage Rooms.
Business Rules	BR-UC41-01: Tour price must be greater than zero. BR-UC41-02: Tour capacity must be greater than zero. BR-UC41-03: Tours with active bookings cannot be deleted. BR-UC41-04: Only active tours can be booked by customers.
Related Use Cases	UC12 View Tour Details UC26 Book Tour UC74 View Tour Manifest
Related Components	Controllers: TourController, TourManagementController Services: TourService, TourScheduleService Repositories / DAO: TourRepository, TourScheduleRepository, TourBookingRepository Entities / Tables: Tours, TourSchedules, TourBookings
Priority	High
Frequency of Use	Medium
Entities Affected	Tours, TourSchedules, TourBookings

2.1.7 UC42 – Manage Menu

Field	Specification
Use Case ID	UC42
Use Case Name	Manage Menu
Feature / Module	Auth, Profiles & Master Data
Primary Actor	Business Manager
Secondary Actors	File Storage Service
Description	As a business manager, I want to manage restaurant menu data so that food items, categories, pricing, and availability remain accurate for restaurant and room service operations.
Preconditions	1. Business Manager is logged into the system. 2. Business Manager has permission to manage menu data.
Postconditions	1. Menu information is created, updated, enabled, disabled, or viewed successfully. 2. Restaurant menu catalog is updated.
Input Data	Menu item name, category, description, price, image, availability status
Output Data	Updated menu item, updated menu catalog, success/error message
Normal Sequence / Flow	Similar to UC40 – Manage Rooms. The managed entity is Menu Item instead of Room.
Alternative Sequence / Flow	Similar to UC40 – Manage Rooms.
Business Rules	BR-UC42-01: Menu item price must be greater than zero. BR-UC42-02: Menu item referenced by historical food orders cannot be physically deleted. BR-UC42-03: Only available menu items can be ordered by customers. BR-UC42-04: Menu item name should be unique within the same category.
Related Use Cases	UC25 Order Food To Room UC66 Update Food Order Status

Related Components	Controllers: MenuController, MenuManagementController Services: MenuService, MenuCategoryService Repositories / DAO: MenuRepository, MenuCategoryRepository, FoodOrderRepository Entities / Tables: MenuItems, MenuCategories, FoodOrders
Priority	High
Frequency of Use	Medium
Entities Affected	MenuItems, MenuCategories, FoodOrders

2.2 Room Reservation & Front Desk

2.2.1 UC04 – Search Available Rooms

Field	Specification
Use Case ID	UC04
Use Case Name	Search Available Rooms
Feature / Module	Room Reservation & Front Desk
Primary Actor	Guest / Customer
Secondary Actors	None
Description	As a guest or customer, I want to search available rooms by date range, guest count, and room type so that I can find a suitable room for my stay.
Preconditions	1. User accesses the room search page. 2. Room and room category data already exist in the system.

Postconditions	1. Available room list is displayed. 2. User can continue to view room details or start room booking.
Input Data	Check-in date, check-out date, number of guests, room type/category, optional price range
Output Data	List of available rooms, room category, price, capacity, room status, search result message
Normal Sequence / Flow	1. User opens the Room Search page. 2. System displays the search form. 3. User enters search criteria. 4. User submits the search request. 5. System checks room availability based on search criteria. 6. System displays available rooms that match the criteria. 7. User selects a room to view details or proceed to booking.
Alternative Sequence / Flow A1 – No room available	At step 5, if no room matches the criteria: 1. System displays “No available rooms found.” 2. System may suggest changing date range, guest count, or room type.
Alternative Sequence / Flow A2 – Invalid search criteria	If check-in/check-out date or guest count is invalid: 1. System rejects the search request. 2. System displays validation messages. 3. User corrects the criteria and searches again.

Exception Flow E1 – System error	If system error occurs: 1. System cannot retrieve room availability. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC04-01: Check-in date must be before check-out date. BR-UC04-02: Number of guests must not exceed room capacity. BR-UC04-03: Rooms with overlapping confirmed bookings must not be displayed as available. BR-UC04-04: Only active rooms can appear in search results.
Related Use Cases	UC06 View Room Details UC22 Book Room
Related Components	Controllers: RoomController, RoomSearchController Services: RoomService, RoomAvailabilityService Repositories / DAO: RoomRepository, BookingRepository, RoomCategoryRepository Entities / Tables: Rooms, RoomCategories, Bookings
Priority	High
Frequency of Use	High
Entities Affected	Rooms, RoomCategories, Bookings

2.2.1 UC22 – Book Room

Field	Specification
Use Case ID	UC22
Use Case Name	Book Room
Feature / Module	Room Reservation & Front Desk
Primary Actor	Customer
Secondary Actors	Payment Gateway
Description	As a customer, I want to book a room so that I can reserve accommodation at the resort for my selected stay period.
Preconditions	1. Customer is logged in. 2. Customer has selected an available room. 3. Check-in/check-out date and guest count are valid.
Postconditions	1. Room booking is created. 2. Booking status is set to Pending Deposit or Confirmed depending on payment status. 3. Room availability is reserved according to booking policy.
Input Data	Customer ID, room ID, check-in date, check-out date, number of guests, special requests
Output Data	Booking record, booking code, booking status, total estimated room cost

Normal Sequence / Flow	1. Customer selects an available room. 2. System displays booking form and booking summary. 3. Customer enters guest and stay information. 4. Customer submits booking request. 5. System validates booking information and room availability. 6. System creates a room booking record. 7. System calculates estimated room cost and required deposit. 8. System displays booking confirmation and payment instruction.
Alternative Sequence / Flow A1 – Room no longer available	At step 5, if the selected room is no longer available: 1. System rejects the booking request. 2. System displays room unavailable message. 3. Customer may return to room search.
Alternative Sequence / Flow A2 – Invalid booking information	If booking information is incomplete or invalid:1. System displays validation messages.2. Customer corrects the information.3. Flow returns to step 4.
Exception Flow E1 – Booking creation error	If system cannot create booking: 1. System cancels the booking process. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC22-01: Double-booking is not allowed for overlapping date ranges. BR-UC22-02: Check-in date must be before check-out date. BR-UC22-03: Guest count must not exceed room capacity.

	BR-UC22-04: Booking must be linked to a customer account. BR-UC22-05: Booking must require deposit payment before being fully confirmed.
Related Use Cases	UC04 Search Available RoomsUC23 Make Deposit PaymentUC27 View Personal InvoiceUC28 View Booking History
Related Components	Controllers: BookingControllerServices: BookingService, RoomAvailabilityService, PricingService Repositories / DAO: BookingRepository, RoomRepository, UserRepository Entities / Tables: Bookings, Rooms, Users, RoomCategories
Priority	Critical
Frequency of Use	High
Entities Affected	Bookings, Rooms, Users, RoomCategories

2.2.3 UC23 – Make Deposit Payment

Field	Specification
Use Case ID	UC23
Use Case Name	Make Deposit Payment
Feature / Module	Room Reservation & Front Desk
Primary Actor	Customer
Secondary Actors	Payment Gateway

Description	As a customer, I want to pay a room deposit so that my room booking can be confirmed.
Preconditions	1. Customer is logged in.2. A booking record exists.3. Booking status is Pending Deposit.4. Payment gateway is available.
Postconditions	1. Deposit payment is recorded.2. Booking status is updated to Confirmed.3. Payment transaction is stored.4. Customer receives booking confirmation.
Input Data	Booking ID, payment method, deposit amount, payment gateway transaction data
Output Data	Payment result, payment transaction record, updated booking status, confirmation message
Normal Sequence / Flow	1. Customer selects Pay Deposit for a booking.2. System displays deposit amount and payment methods.3. Customer selects payment method and confirms payment.4. System redirects or sends payment request to payment gateway.5. Payment gateway processes the transaction.6. System receives payment result.7. System records payment transaction.8. System updates booking status to Confirmed.9. System displays payment success and booking confirmation.
Alternative Sequence / Flow A1 – Payment failed	At step 6, if payment fails:1. System records failed payment attempt if applicable.2. Booking remains Pending Deposit.3. System displays payment failed message.4. Customer may retry payment.
Alternative Sequence / Flow A2 – Customer cancels payment	If customer cancels payment at gateway:1. System returns customer to booking page.2. Booking remains Pending Deposit.3. System displays payment cancelled message.
Exception Flow E1 – Payment callback error	If system cannot verify payment result:1. System keeps booking in Pending Verification or Pending Deposit status.2. System records the payment verification error.3. System displays pending payment message.

Business Rules	BR-UC23-01: Deposit payment must be linked to a valid booking.BR-UC23-02: Only successful payment can confirm a booking.BR-UC23-03: Payment transaction must be recorded for audit and billing.BR-UC23-04: Deposit refund follows cancellation policy.
Related Use Cases	UC22 Book RoomUC54 Generate Consolidated InvoiceUC55 Process Final Payment
Related Components	Controllers: PaymentController, BookingControllerServices: PaymentService, PaymentGatewayService, BookingServiceRepositories / DAO: PaymentRepository, BookingRepositoryEntities / Tables: Payments, Bookings, Refunds
Priority	Critical
Frequency of Use	High
Entities Affected	Payments, Bookings, Refunds

2.2.4 UC51 – Perform Customer Check-in

Field	Specification
Use Case ID	UC51
Use Case Name	Perform Customer Check-in
Feature / Module	Room Reservation & Front Desk
Primary Actor	Receptionist
Secondary Actors	Customer

Description	As a receptionist, I want to check guests in so that they can begin their stay after their booking and identity information are verified.
Preconditions	1. Receptionist is logged in.2. Customer has a confirmed booking.3. Customer arrives at the resort.4. Booking is not already checked in or cancelled.
Postconditions	1. Booking status is updated to Checked-in.2. A physical room is assigned to the customer.3. Room status is updated to Occupied.4. Guest stay information is ready for stay registration.
Input Data	Booking code, customer identity information, assigned room number, check-in confirmation
Output Data	Updated booking status, assigned room, updated room status, check-in result message
Normal Sequence / Flow	1. Receptionist opens the Expected Arrival or Booking Management page.2. Receptionist searches and selects the customer booking.3. System displays booking and customer information.4. Receptionist verifies customer identity and booking details.5. Receptionist assigns an available physical room.6. Receptionist confirms check-in.7. System updates booking status to Checked-in.8. System updates room status to Occupied.9. System displays check-in success message.
Alternative Sequence / Flow A1 – Booking not found	If booking cannot be found:1. System displays booking not found message.2. Receptionist may search again using different information.
Alternative Sequence / Flow A2 – Identity information does not match	If customer identity does not match booking information:1. Receptionist stops the check-in process.2. System keeps booking status unchanged.3. Receptionist may update or verify customer information according to policy.

Alternative Sequence / Flow A3 – No available physical room	If no physical room is available for assignment:1. System displays unavailable room message.2. Receptionist may assign another suitable room or escalate to manager.
Exception Flow E1 – System error	If system error occurs during check-in:1. System cancels check-in update.2. Booking and room status remain unchanged.3. System records the error.
Business Rules	BR-UC51-01: Only confirmed bookings can be checked in.BR-UC51-02: A booking cannot be checked in more than once.BR-UC51-03: A physical room must be assigned during check-in.BR-UC51-04: Assigned room must be available and suitable for the booking.BR-UC51-05: Room status must become Occupied after successful check-in.
Related Use Cases	UC50 View Daily Arrival/Departure ListUC52 Submit Stay RegistrationUC56 Update Room Operational Status
Related Components	Controllers: ReceptionistController, CheckInControllerServices: CheckInService, BookingService, RoomStatusServiceRepositories / DAO: BookingRepository, RoomRepository, UserProfileRepositoryEntities / Tables: Bookings, Rooms, Users, UserProfiles
Priority	Critical
Frequency of Use	High
Entities Affected	Bookings, Rooms, Users, UserProfiles

2.2.5 UC52 – Submit Stay Registration

Field	Specification
Use Case ID	UC52

Use Case Name	Submit Stay Registration
Feature / Module	Room Reservation & Front Desk
Primary Actor	Receptionist
Secondary Actors	Customer
Description	As a receptionist, I want to submit guest stay registration information so that the resort can comply with temporary residence regulations.
Preconditions	1. Receptionist is logged in.2. Customer has a confirmed booking or has completed check-in.3. Customer identity information is available or provided at front desk.
Postconditions	1. Guest stay registration information is recorded.2. Stay registration is linked to the booking.3. Registration data can be used for temporary residence reporting.
Input Data	Booking ID, guest full name, date of birth, nationality, CCCD/Passport number, phone number, address, check-in date, assigned room
Output Data	Stay registration record, registration status, success/error message
Normal Sequence / Flow	1. Receptionist opens the Stay Registration form from the booking/check-in screen.2. System displays booking and guest information.3. Receptionist enters or confirms required stay registration information.4. Receptionist submits the registration form.5. System validates required information.6. System saves stay registration data.7. System displays registration success message.
Alternative Sequence / Flow A1 – Missing required information	At step 5, if required information is missing:1. System rejects submission.2. System displays missing field messages.3. Receptionist updates the information and submits again.

Alternative Sequence / Flow A2 – Invalid identity document information	If CCCD/Passport information is invalid:1. System displays validation message.2. Receptionist corrects or updates guest identity information.
Exception Flow E1 – Save registration error	If system cannot save registration data:1. System displays a general error message.2. Registration status remains incomplete.3. System records the error.
Business Rules	BR-UC52-01: Stay registration must include required guest identity fields.BR-UC52-02: Stay registration must be linked to a valid booking.BR-UC52-03: Sensitive identity information must be stored securely.BR-UC52-04: Registration data should be available for temporary residence report export.
Related Use Cases	UC17 Update CCCD/PassportUC51 Perform Customer Check-inUC58 Export Temporary Residence Report
Related Components	Controllers: StayRegistrationController, ReceptionistControllerServices: StayRegistrationService, GuestIdentityService, EncryptionServiceRepositories / DAO: StayRegistrationRepository, BookingRepository, UserProfileRepositoryEntities / Tables: StayRegistrations, Bookings, UserProfiles, Rooms
Priority	High
Frequency of Use	High
Entities Affected	StayRegistrations, Bookings, UserProfiles, Rooms

2.2.6 UC53 – Perform Customer Check-out

Field	Specification

Use Case ID	UC53
Use Case Name	Perform Customer Check-out
Feature / Module	Room Reservation & Front Desk
Primary Actor	Receptionist
Secondary Actors	Customer, Payment Gateway
Description	As a receptionist, I want to check guests out so that their stay can be completed after all charges are reviewed and settled.
Preconditions	1. Receptionist is logged in.2. Customer is currently checked in.3. Customer booking has an active guest folio.4. Room, food, and tour charges are available for billing.
Postconditions	1. Consolidated invoice is generated or confirmed.2. Final payment is completed.3. Booking status is updated to Checked-out.4. Room status is updated to Needs Cleaning.5. Guest folio is closed.
Input Data	Booking ID, folio information, payment method, checkout confirmation
Output Data	Checked-out booking, paid invoice, closed folio, updated room status, checkout result message
Normal Sequence / Flow	1. Receptionist opens the Check-out page.2. Receptionist searches and selects checked-in booking.3. System displays booking, room, food, tour, and folio charges.4. Receptionist reviews outstanding charges with customer.5. System generates or displays consolidated invoice.6. Receptionist processes final payment.7. System marks invoice as paid.8. Receptionist confirms check-out.9. System updates booking status to Checked-out.10. System updates room status to Needs Cleaning.11. System closes the guest folio and displays success message.

Alternative Sequence / Flow A1 – Unpaid charges exist	If unpaid charges are detected:1. System prevents checkout completion.2. System displays outstanding charge details.3. Receptionist must process payment before continuing.
Alternative Sequence / Flow A2 – Customer disputes charge	If customer disputes a charge:1. Receptionist reviews charge details.2. System allows authorized adjustment if permitted.3. Invoice is recalculated before payment.
Exception Flow E1 – Payment failure	If final payment fails:1. System does not complete checkout.2. Invoice remains unpaid.3. Booking remains checked-in.4. System displays payment failure message.
Business Rules	BR-UC53-01: Customer cannot be checked out if invoice is unpaid.BR-UC53-02: All unpaid room, food, and tour charges must be included in consolidated invoice.BR-UC53-03: Guest folio must be closed after successful checkout.BR-UC53-04: Room status must become Needs Cleaning after checkout.
Related Use Cases	UC54 Generate Consolidated InvoiceUC55 Process Final PaymentUC56 Update Room Operational Status
Related Components	Controllers: CheckOutController, ReceptionistControllerServices: CheckOutService, InvoiceService, PaymentService, FolioService, RoomStatusServiceRepositories / DAO: BookingRepository, InvoiceRepository, PaymentRepository, GuestFolioRepository, RoomRepositoryEntities / Tables: Bookings, GuestFolios, Invoices, Payments, Rooms
Priority	Critical
Frequency of Use	High
Entities Affected	Bookings, GuestFolios, Invoices, Payments, Rooms

2.2.7 UC56 – Update Room Operational Status

Field	Specification
Use Case ID	UC56
Use Case Name	Update Room Operational Status
Feature / Module	Room Reservation & Front Desk
Primary Actor	Receptionist
Secondary Actors	Housekeeping Staff, Maintenance Staff
Description	As a receptionist, I want to update room operational statuses so that room availability and operational conditions remain accurate.
Preconditions	1. Receptionist is logged in.2. Room exists in the system.3. Receptionist has permission to update room status.
Postconditions	1. Room operational status is updated.2. Updated status is reflected in room search, check-in, housekeeping, or maintenance workflows.3. Status change activity is recorded.
Input Data	Room ID, new room status, status change reason, optional note
Output Data	Updated room status, status update result message

Normal Sequence / Flow	1. Receptionist opens Room Status Management page.2. System displays room list with current statuses.3. Receptionist selects a room.4. Receptionist selects a new operational status.5. Receptionist submits the status update.6. System validates the status transition.7. System updates room status.8. System displays update success message.
Alternative Sequence / Flow A1 – Invalid status transition	If selected status transition is not allowed:1. System rejects the update.2. System displays invalid status transition message.3. Receptionist selects another status.
Alternative Sequence / Flow A2 – Room has active booking	If room status change conflicts with active booking:1. System prevents or warns about the update.2. Receptionist must resolve booking conflict before updating.
Exception Flow E1 – System error	If system cannot update room status:1. Room status remains unchanged.2. System displays a general error message.3. System records the error.
Business Rules	BR-UC56-01: Room status must follow allowed room state transition rules.BR-UC56-02: Occupied room cannot be marked Available directly without checkout/cleaning process.BR-UC56-03: Rooms in Maintenance status cannot be booked.BR-UC56-04: Rooms marked Needs Cleaning should appear in housekeeping task flow.
Related Use Cases	UC51 Perform Customer Check-inUC53 Perform Customer Check-outUC60 View Room Cleaning TasksUC63 Report Room Issues
Related Components	Controllers: RoomStatusController, ReceptionistControllerServices: RoomStatusService, HousekeepingTaskService, MaintenanceTaskServiceRepositories / DAO: RoomRepository, CleaningTaskRepository, MaintenanceTaskRepository, ActivityLogRepositoryEntities / Tables: Rooms, CleaningTasks, MaintenanceTasks, ActivityLogs
Priority	High

Frequency of Use	High
Entities Affected	Rooms, CleaningTasks, MaintenanceTasks, ActivityLogs

2.3 Restaurant POS & Room Service

2.3.1 UC24 – Reserve Restaurant Table

Field	Specification
Use Case ID	UC24
Use Case Name	Reserve Restaurant Table
Feature / Module	Restaurant POS & Room Service
Primary Actor	Customer
Secondary Actors	Receptionist
Description	As a customer, I want to reserve a restaurant table so that I can dine at the resort restaurant at my preferred time.
Preconditions	1. Customer is logged in. 2. Restaurant table information exists in the system. 3. Restaurant is operating during the requested reservation time.
Postconditions	1. Restaurant reservation is created. 2. Reserved table is allocated for the selected time slot. 3. Reservation information is available to restaurant staff.
Input Data	Customer ID, reservation date, reservation time, number of guests, special requests
Output Data	Restaurant reservation record, reservation code, reservation status
Normal Sequence / Flow	1. Customer opens Restaurant Reservation page. 2. System displays available reservation information. 3. Customer enters reservation details. 4. Customer submits reservation request. 5. System checks table availability. 6. System creates reservation.

	7. System displays reservation confirmation.
Alternative Sequence / Flow A1 – No table available	At step 5, if no suitable table is available: 1. System rejects the reservation request. 2. System displays unavailable table message. 3. Customer may select another time slot.
Alternative Sequence / Flow A2 – Invalid reservation information	System displays validation messages and requests correction.
Exception Flow E1 – System error	Reservation cannot be created and system displays an error message.
Business Rules	BR-UC24-01: Reservation time must be within restaurant operating hours. BR-UC24-02: Guest count must not exceed table capacity. BR-UC24-03: Table cannot be double-booked for the same time slot.
Related Use Cases	UC09 View Restaurant Menu
Related Components	Controllers: RestaurantReservationController Services: RestaurantReservationService, TableAvailabilityService Repositories / DAO: RestaurantReservationRepository, TableRepository Entities / Tables: RestaurantReservations, Tables
Priority	Medium
Frequency of Use	Medium
Entities Affected	RestaurantReservations, Tables

2.3.2 UC25 – Order Food To Room

Field	Specification
Use Case ID	UC25
Use Case Name	Order Food To Room
Feature / Module	Restaurant POS & Room Service

Primary Actor	Customer
Secondary Actors	Kitchen Staff
Description	As a customer, I want to order food to my room so that I can enjoy room service during my stay.
Preconditions	1. Customer is checked in.2. Customer has an active room assignment.3. Menu items are available.
Postconditions	1. Food order is created.2. Order is sent to restaurant operations.3. Order status is initialized for processing.
Input Data	Booking Id, ordered menu items, quantity > 0, special requests
Output Data	Food order record, order number, order status, order total amount
Normal Sequence / Flow	1. Customer opens the room service menu. 2. System displays available menu items. 3. Customer selects food items and quantities. 4. Customer submits the order. 5. System validates the order information. 6. System creates the food order. 7. System sends the order to restaurant operations. 8. System displays order confirmation.
Alternative Sequence / Flow A1 – Menu item unavailable	System informs customer that selected item is unavailable and requests another selection.
Alternative Sequence / Flow A2 – Invalid room information	System rejects the order and displays validation message.
Exception Flow E1 – System error	Order is not created and system displays an error message.
Business Rules	BR-UC25-01: Only checked-in customers may use room service. BR-UC25-02: Orders must be associated with an occupied room. BR-UC25-03: Only available menu items can be ordered. BR-UC25-04: The unit price of each menu item must be snapshotted at the time of order creation.
Related Use Cases	UC66 Update Food Order StatusUC69 Charge Food Bill To Room

Related Components	Controllers: RoomServiceOrderController Services: FoodOrderService Repositories / DAO: FoodOrderRepository, MenuRepository, RoomRepository Entities / Tables: FoodOrders, FoodOrderItems, MenuItems, Rooms
Priority	High
Frequency of Use	High
Entities Affected	FoodOrders, FoodOrderItems, MenuItems, Rooms

2.3.3 UC66 – Update Food Order Status

Field	Specification
Use Case ID	UC66
Use Case Name	Update Food Order Status
Feature / Module	Restaurant POS & Room Service
Primary Actor	Restaurant Staff
Secondary Actors	Kitchen Staff
Description	As restaurant staff, I want to update food order status so that order progress can be tracked accurately.
Preconditions	1. Food order exists. 2. Staff has permission to manage food orders.
Postconditions	1. Food order status is updated. 2. Updated status is visible to relevant staff.
Input Data	Order ID, new order status, optional note
Output Data	Updated food order status

Normal Sequence / Flow	1. Staff opens Food Order Management. 2. System displays food orders. 3. Staff selects an order. 4. Staff updates the order status. 5. System validates the status update. 6. System saves the new status. 7. System displays update result.
Alternative Sequence / Flow A1 – Invalid status transition	System rejects the update and displays a warning message.
Exception Flow E1 – System error	System cannot save the status update and displays an error message.
Business Rules	BR-UC66-01: Order status must follow allowed order lifecycle. BR-UC66-02: Completed orders cannot return to a previous status. BR-UC66-03: The allowed order status transition is: Pending ➔ Cooking ➔ Delivered.
Related Use Cases	UC25 Order Food To Room UC73 Update KOT Status
Related Components	Controllers: FoodOrderController Services: FoodOrderService Repositories / DAO: FoodOrderRepository Entities / Tables: FoodOrders
Priority	High
Frequency of Use	High
Entities Affected	FoodOrders

2.3.4 UC69 – Charge Food Bill To Room

Field	Specification
Use Case ID	UC69
Use Case Name	Charge Food Bill To Room
Feature / Module	Restaurant POS & Room Service
Primary Actor	Restaurant Staff
Secondary Actors	Receptionist

Description	As restaurant staff, I want to charge a food order to a guest room so that the charge is included in the guest's final invoice.
Preconditions	1. Food order exists.2. Food order has been completed or delivered.3. Room is occupied by a checked-in guest.
Postconditions	1. Food charge is linked to guest folio.2. Charge becomes available for consolidated billing.
Input Data	Food order ID, room number
Output Data	Posted food charge, updated guest folio
Normal Sequence / Flow	1. Staff opens completed food order information. 2. Staff selects Charge To Room. 3. System displays room information. 4. Staff confirms the room assignment. 5. System validates room occupancy. 6. System posts the charge to the guest folio. 7. System displays successful charge confirmation.
Alternative Sequence / Flow A1 – Room not occupied	System rejects the charge request and displays warning message.
Alternative Sequence / Flow A2 – Invalid room information	System requests room information correction. The food order has already been charged: 1. System detects the order was previously charged to the room. 2. System displays an error message "Order already charged to room" and prevents the action.
Exception Flow E1 – Posting error	Charge is not posted and system displays an error message.
Business Rules	BR-UC69-01: Only completed orders can be charged to a room. BR-UC69-02: Room must belong to a currently checked-in guest. BR-UC69-03: Food charge must appear in consolidated billing. BR-UC69-04: A completed food order can only be charged to a room ONCE. The system must prevent duplicate charges for the same order.
Related Use Cases	UC25 Order Food To RoomUC54 Generate Consolidated Invoice

Related Components	Controllers: FoodOrdersController Services: FoodOrderService, GuestFolioService Repositories / DAO: FoodOrderRepository, GuestFolioRepository Entities / Tables: FoodOrders, GuestFolios, ChargeItems
Priority	High
Frequency of Use	High
Entities Affected	FoodOrders, GuestFolios, FolioTransactions

2.3.5 UC73 – Update KOT Status

Field	Specification
Use Case ID	UC73
Use Case Name	Update KOT Status
Feature / Module	Restaurant POS & Room Service
Primary Actor	Kitchen Staff
Secondary Actors	Restaurant Staff
Description	As kitchen staff, I want to update Kitchen Order Ticket (KOT) status so that restaurant staff can track food preparation progress.
Preconditions	1. KOT exists. 2. Kitchen staff is authenticated.
Postconditions	1. KOT status is updated. 2. Updated status is visible to restaurant operations.
Input Data	KOT ID, new KOT status
Output Data	Updated KOT status
Normal Sequence / Flow	1. Kitchen staff opens KOT management screen. 2. System displays active KOTs. 3. Kitchen staff selects a KOT. 4. Kitchen staff updates KOT status. 5. System validates the status update. 6. System saves the new status. 7. System displays update result.

Alternative Sequence / Flow A1 – Invalid status update	System rejects the update and displays warning message.
Exception Flow E1 – System error	System cannot save KOT status and displays error message.
Business Rules	BR-UC73-01: KOT status must follow the kitchen workflow. BR-UC73-02: Completed KOTs cannot return to previous statuses. BR-UC73-03: The allowed Kitchen Order Ticket (KOT) status transition is: Pending ➔ Cooking ➔ Ready ➔ Completed.
Related Use Cases	UC25 Order Food To RoomUC66 Update Food Order Status
Related Components	Controllers: KOTController Services: KOTService Repositories / DAO: KOTRepository Entities / Tables: KOTs, FoodOrders
Priority	High
Frequency of Use	High
Entities Affected	KOTs, FoodOrders

2.4 Tour Booking & Feedback System

2.4.1 UC26 – Book Tour

Field	Specification
Use Case ID	UC26
Use Case Name	Book Tour
Feature / Module	Tour Booking & Feedback System
Primary Actor	Customer
Secondary Actors	Tour Guide
Description	As a customer, I want to book a tour so that I can participate in resort activities during my stay.
Preconditions	1. Customer is logged in. 2. Tour information and tour schedule exist in the system.

	3. Selected tour schedule is active and available.
Postconditions	1. Tour booking is created. 2. Number of available tour slots is updated. 3. Tour booking can be included in the guest's consolidated invoice.
Input Data	Customer ID, tour schedule ID, number of participants, booking note
Output Data	Tour booking record, tour booking status, booking confirmation message
Normal Sequence / Flow	1. Customer opens Tour List or Tour Detail page. 2. System displays tour information and available schedules. 3. Customer selects a tour schedule and number of participants. 4. Customer submits the tour booking request. 5. System validates tour availability and participant count. 6. System creates the tour booking. 7. System displays tour booking confirmation.
Alternative Sequence / Flow A1 – Tour schedule unavailable	At step 5, if the selected schedule is unavailable: 1. System rejects the booking request. 2. System displays unavailable schedule message. 3. Customer may select another schedule.
Alternative Sequence / Flow A2 – Participant count exceeds available slots	At step 5, if participant count exceeds available slots: 1. System rejects the booking request. 2. System displays available slot information. 3. Customer may reduce participant count or select another schedule.
Exception Flow E1 – System error	If system error occurs during booking: 1. System does not create the tour booking. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC26-01: Customer can only book active tour schedules. BR-UC26-02: Number of participants must be greater than zero. BR-UC26-03: Number of participants must not exceed available capacity. BR-UC26-04: Tour booking must be linked to the customer or active booking/folio if applicable. BR-UC26-05: Tour charges must be available for consolidated billing.
Related Use Cases	UC12 View Tour Details UC54 Generate Consolidated Invoice UC29 Submit Reviews

Related Components	Controllers: TourBookingController, TourController Services: TourBookingService, TourScheduleService Repositories / DAO: TourBookingRepository, TourScheduleRepository, TourRepository Entities / Tables: TourBookings, TourSchedules, Tours, Users
Priority	High
Frequency of Use	Medium
Entities Affected	TourBookings, TourSchedules, Tours, Users

2.4.2 UC29 – Submit Reviews

Field	Specification
Use Case ID	UC29
Use Case Name	Submit Reviews
Feature / Module	Tour Booking & Feedback System
Primary Actor	Customer
Secondary Actors	Business Manager
Description	As a customer, I want to submit a rating and written review so that I can share my experience after using resort services.
Preconditions	1. Customer is logged in. 2. Customer has completed a stay, food order, or tour service depending on review type. 3. Customer has not violated review submission rules.
Postconditions	1. Review is submitted successfully. 2. Review status is set according to moderation policy. 3. Review becomes available for manager moderation or public display.

Input Data	Customer ID, service type, related service ID, star rating, review content
Output Data	Review record, review status, submission result message
Normal Sequence / Flow	1. Customer opens Review & Feedback page. 2. System displays eligible services for review. 3. Customer selects a service to review. 4. Customer enters star rating and written review. 5. Customer submits the review. 6. System validates review information. 7. System saves the review.8. System displays review submission result.
Alternative Sequence / Flow A1 – Customer is not eligible to review	At step 2 or step 6, if customer has not completed the service: 1. System does not allow review submission. 2. System displays ineligible review message.
Alternative Sequence / Flow A2 – Invalid review content	At step 6, if rating or content is invalid: 1. System rejects the review. 2. System displays validation message. 3. Customer corrects the review and submits again.
Exception Flow E1 – System error	If system error occurs during review submission: 1. System does not save the review. 2. System displays a general error message. 3. System records the error.
Business Rules	BR-UC29-01: Star rating must be within the allowed range, such as 1 to 5 stars. BR-UC29-02: Customer can only review services they have used or completed. BR-UC29-03: Review content must not be empty if written review is required. BR-UC29-04: Review may require manager approval before public display.
Related Use Cases	UC14 View Reviews UC26 Book Tour UC48 Approve/Hide Review
Related Components	Controllers: ReviewControllerServices: ReviewService Repositories / DAO: ReviewRepository, BookingRepository, TourBookingRepository, FoodOrderRepository

	Entities / Tables: Reviews, Users, Bookings, TourBookings, FoodOrders
Priority	Medium
Frequency of Use	Medium
Entities Affected	Reviews, Users, Bookings, TourBookings, FoodOrders

2.4.3 UC48 – Approve/Hide Review

Field	Specification
Use Case ID	UC48
Use Case Name	Approve/Hide Review
Feature / Module	Tour Booking & Feedback System
Primary Actor	Business Manager
Secondary Actors	Customer
Description	As a business manager, I want to approve or hide customer reviews so that only appropriate feedback is displayed on the public portal.
Preconditions	1. Business Manager is logged in. 2. Business Manager has permission to moderate reviews. 3. Customer review exists in the system.
Postconditions	1. Review display status is updated. 2. Public review list is updated according to moderation decision. 3. Moderation activity is recorded.
Input Data	Review ID, moderation action, optional moderation note
Output Data	Updated review status, moderation result message
Normal Sequence / Flow	1. Business Manager opens Customer Review Management page. 2. System displays customer reviews. 3. Business Manager selects a review. 4. Business Manager chooses Approve or Hide. 5. System updates the review status. 6. System displays moderation result.

Alternative Sequence / Flow A1 – Review already moderated	If the selected review has already been moderated: 1. System displays current review status. 2. Business Manager may keep or change the moderation decision if permitted.
Alternative Sequence / Flow A2 – Review not found	If the review no longer exists: 1. System displays review not found message. 2. Review status remains unchanged.
Exception Flow E1 – System error	If system cannot update review status: 1. System displays a general error message. 2. Review status remains unchanged.3. System records the error.
Business Rules	BR-UC48-01: Only Business Manager can moderate customer reviews. BR-UC48-02: Hidden reviews must not be displayed on the public portal. BR-UC48-03: Approved reviews can be displayed publicly. BR-UC48-04: Moderation changes should be recorded for tracking.
Related Use Cases	UC29 Submit Reviews UC14 View Reviews
Related Components	Controllers: ReviewManagementControlle rServices: ReviewModerationService, ReviewService Repositories / DAO: ReviewRepository, ActivityLogRepository Entities / Tables: Reviews, ActivityLogs
Priority	Medium
Frequency of Use	Medium
Entities Affected	Reviews, ActivityLogs

…

3. Functional Requirements

[Provide descriptions about the system's functions/screens. The functions/screens are grouped by the system features, and even sub-features if needed. For the screens, you need to provide the screen layouts (mock-up screens) and relevant specifications if needed]

3.1 Feature Name1

3.1.1 SubFeature Name1.1

3.1.1.1 Screen/Function Name1

[Content #1: UI layout (Mockup screen prototype)]

[Content #2: brief descriptions of the screen/function, mapped to the relevant use cases]

[Content #3: provide further descriptions for the screen's components/fields using table format below]

Field Name	Description
Field Name1	Field description: data type min/max length or value, initial data, etc.
Field Name2	…
Field Group-Name1	
Field Name3	…
Field Name4	…
Field Group-Name2	
…	…

3.1.1.2 Screen/Function Name2

…

3.1.2 SubFeature Name1.2

…

3.2 User Authentication

3.2.1 User Register

…

3.2.2 User Login

…

3.2.3 Password Reset

…

3.3 System Administration

3.3.1 Master Data

3.3.1.1 Setting List

This screen allows the Administrator to:

- View Setting List: view list of current master data.
- Filter Setting List: filter master data by data types, statuses
- Search Settings: enter keyword(s) to search master data by their names or values
- Sort Setting List: sort master data list (ascending, descending) by clicking column headers

On the screen, s/he can also

- Activate/Deactivate Setting: change status of a specific inactive/active master data
- Choose to go to the Setting Details screens for adding new or updating an existing master data by clicking the New Setting or Edit link.

Field Description

Field Name	Description
(1)	Initial values: all the active setting names with null or blank type Hover the mouse to show the field name: “Setting Type”
(2)	Initial values: All Statuses, Active, Inactive (default value “All Status”) Hover the mouse to show the field name: “Setting Status”
(3)	The change-status action is Activate or Deactivate depending on the current status of the relevant setting (Inactive or Active, respectively).

3.3.1.2 Setting Details

This screen allows the Administrator to:

- Add New Setting: add new master data.
- Update Setting Details: update details of a specific master data

Field Description

Field Name	Description
Name	Data type: non-digit string, max length of 20 characters
Type	Initial data values: all active setting names (with null or blank type)
Value	Data type: any string, max length of 100 characters
Priority	Data type: a positive integer
Description	Data type: any string, max length of 200 characters

3.3.2 User Management

3.3.2.1 User List

…

3.3.2.2 User Details

…

4. Non-Functional Requirements

4.1 External Interfaces

[This section provides information to ensure that the system will communicate properly with users and with external hardware or software/system elements.]

4.2 Quality Attributes

[List all the required system characteristics (quality attributes) specification. Some of the possible attributes are provided with the guide/descriptions are mentioned here]

4.2.1 Usability

[This section includes all those requirements that affect usability. For example, specify the required training time for a normal user and a power user to become productive at particular operations specify measurable task times for typical tasks or base the new system's usability requirements on other systems that the users know and like specify requirement to conform to common usability standards, such as IBM's CUA standards Microsoft's GUI standards]

4.2.2 Performance

[The system's performance characteristics are outlined in this section. Include specific response times. Where applicable, reference related Use Cases by name.

Response time for a transaction (average, maximum)

Throughput, for example, transactions per second

Capacity, for example, the number of customers or transactions the system can accommodate

Resource utilization, such as memory, disk, communications, and so forth.]

4.2.3 …

5. Requirement Appendix

[Provide business rules, common requirements, or other extra requirements information here]

5.1 Business Rules

[Provide common business rules that you must follow. The information can be provided in the table format as the sample below]

ID	Rule Definition
BR-01	Delivery time windows are 15 minutes, beginning on each quarter hour.
BR-02	Deliveries must be completed between 10:00 A.M. and 2:00 P.M. local time, inclusive.
BR-03	All meals in a single order must be delivered to the same location.
BR-04	All meals in a single order must be paid for by using the same payment method.

BR-11	If an order is to be delivered, the patron must pay by payroll deduction.
BR-12	Order price is calculated as the sum of each food item price times the quantity of that food item ordered, plus applicable sales tax, plus a delivery charge if a meal is delivered outside the free delivery zone.
BR-24	Only cafeteria employees who are designated as Menu Managers by the Cafeteria Manager can create, modify, or delete cafeteria menus.
BR-33	Network transmissions that involve financial information or personally identifiable information require 256-bit encryption.
BR-86	Only regular employees can register for payroll deduction for any company purchase.
BR-88	An employee can register for payroll deduction payment of cafeteria meals if no more than 40 percent of his gross pay is currently being deducted for other reasons.

5.2 System Messages

#	Message code	Message Type	Context	Content
1	MSG01	In line	There is not any search result	*No search results.*
2	MSG02	In red, under the text box	Input-required fields are empty	*The * field is required.*
3	MSG03	Toast message	Updating asset(s) information successfully	*Update asset(s) successfully.*
4	MSG04	Toast message	Adding new asset successfully	*Add asset successfully.*
5	MSG05	Toast message	Confirming email of asset hand-over is sent successfully	*A confirmation email has been sent to {email_address}.*
6	MSG06	Toast message	Resetting asset information successfully	*Return asset(s) successfully.*
7	MSG07	Toast message	Deleting asset information successfully	*Delete asset(s) successfully.*
8	MSG08	In red, under the text box	Input value length > max length	*Exceed max length of {max_length}.*

9	MSG09	In line	Username or password is not correct when clicking sign-in	*Incorrrect user name or password. Please check again.*
10	..			

5.3 Other Requirements…